Qlik Estate Rationalisation

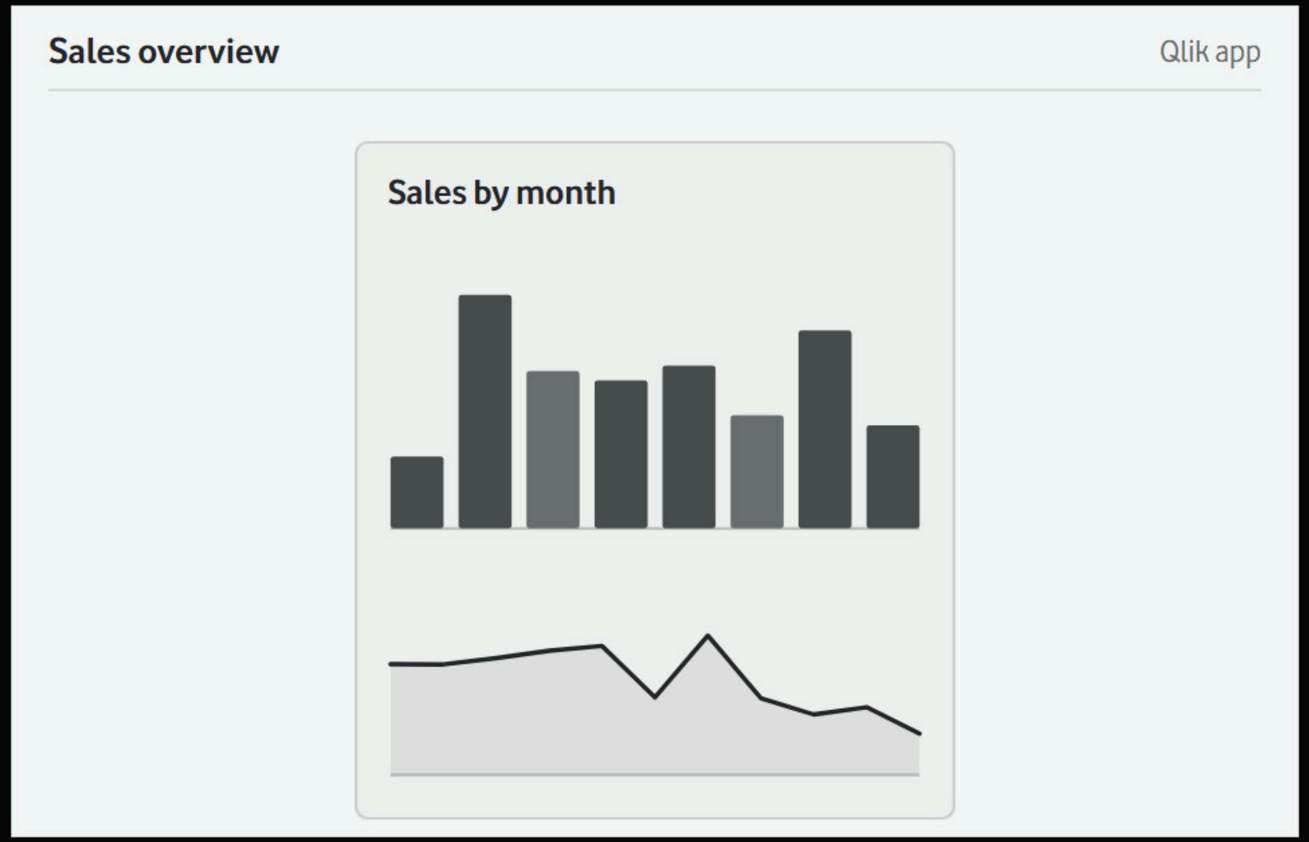
Building Qlik Lineage from the Ground Up

Sebastián Cascardo Andrés • Qlik Specialist Technical Lead • AIDA

```
20261012T060000.040+0200  0021  // ---- Extract: sales orders ----
20261012T060000.041+0200  0022  LIB CONNECT TO 'SALES_DB'
20261012T060000.420+0200  0024  SQL SELECT * FROM dbo.SalesOrders
20261012T060012.310+0200  0026  STORE SalesOrders INTO [lib://SalesContainer/QVD/sales_orders.qvd] (qvd)
20261012T060013.103+0200  0033  SQL SELECT CustomerID, Segment, Region FROM dbo.Customers
20261012T060014.990+0200  0035  STORE Customers INTO [lib://SalesContainer/QVD/customers.qvd] (qvd)
20261012T060015.402+0200        Execution finished.

20261012T070200.035+0200  0018  // ---- Data model ----
20261012T070200.036+0200  0019  Sales:
20261012T070200.037+0200  0020  LOAD * FROM [lib://SalesContainer/QVD/sales_orders.qvd] (qvd)
20261012T070201.420+0200  0023  Customers:
20261012T070201.421+0200  0024  LOAD * FROM [lib://SalesContainer/QVD/customers.qvd] (qvd)
20261012T070201.540+0200  0027  Calendar:
20261012T070201.541+0200  0028  LOAD * FROM [lib://MasterContainer/QVD/calendar.qvd] (qvd)
20261012T070201.610+0200        Creating search index
20261012T070202.204+0200        Search index creation completed successfully
20261012T070202.215+0200        App saved
```

Qlik in 90 seconds



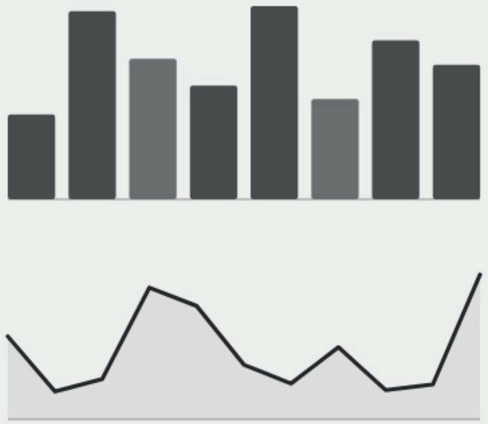
Qlik in 90 seconds

App = a dashboard + the script that loads its data

Sales overview Qlik app

Dashboard

Sales by month



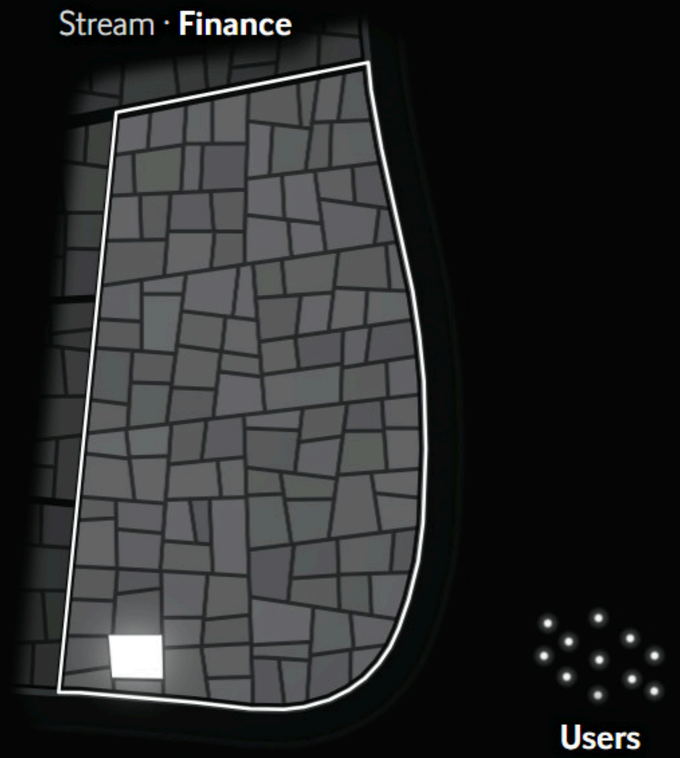
Script

```
12 LIB CONNECT TO 'SALES_DB';
13 Sales:
14 SQL SELECT Region, Month, Amount
15 FROM orders;
```

Reload

the app runs its script
and refreshes its data

Qlik in 90 seconds

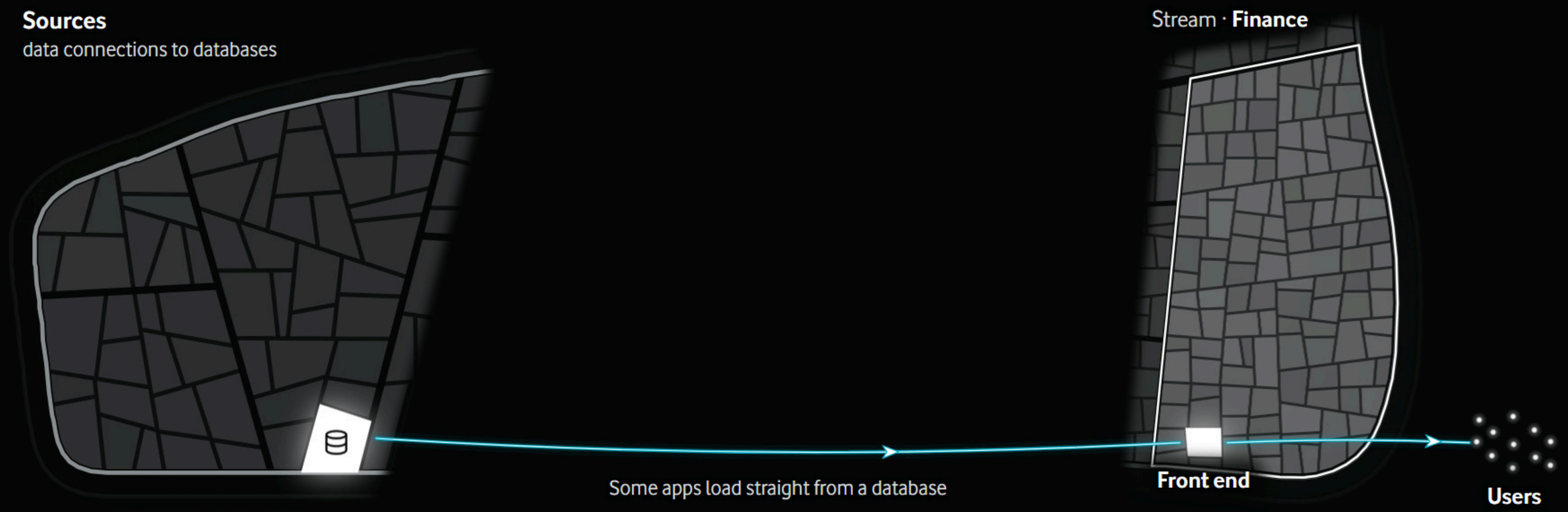


App a dashboard + the script that loads its data

Stream a folder in the Qlik hub, where users find their apps

Reload the app runs its script and refreshes its data

Qlik in 90 seconds



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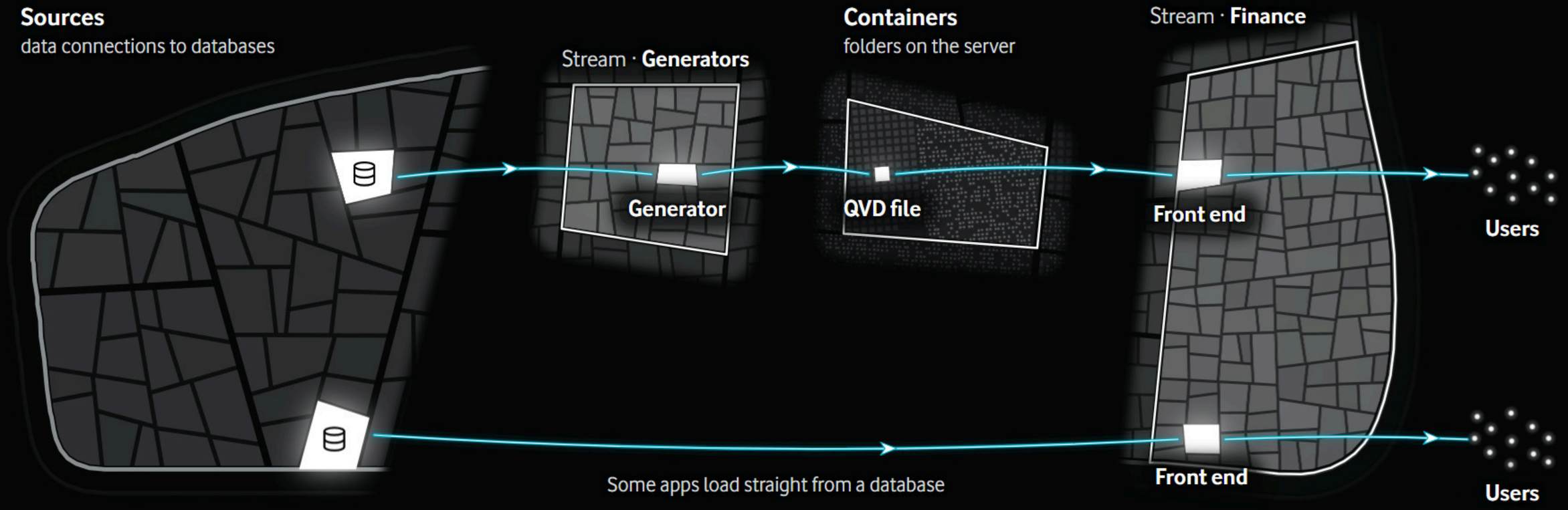


App a dashboard + the script that loads its data

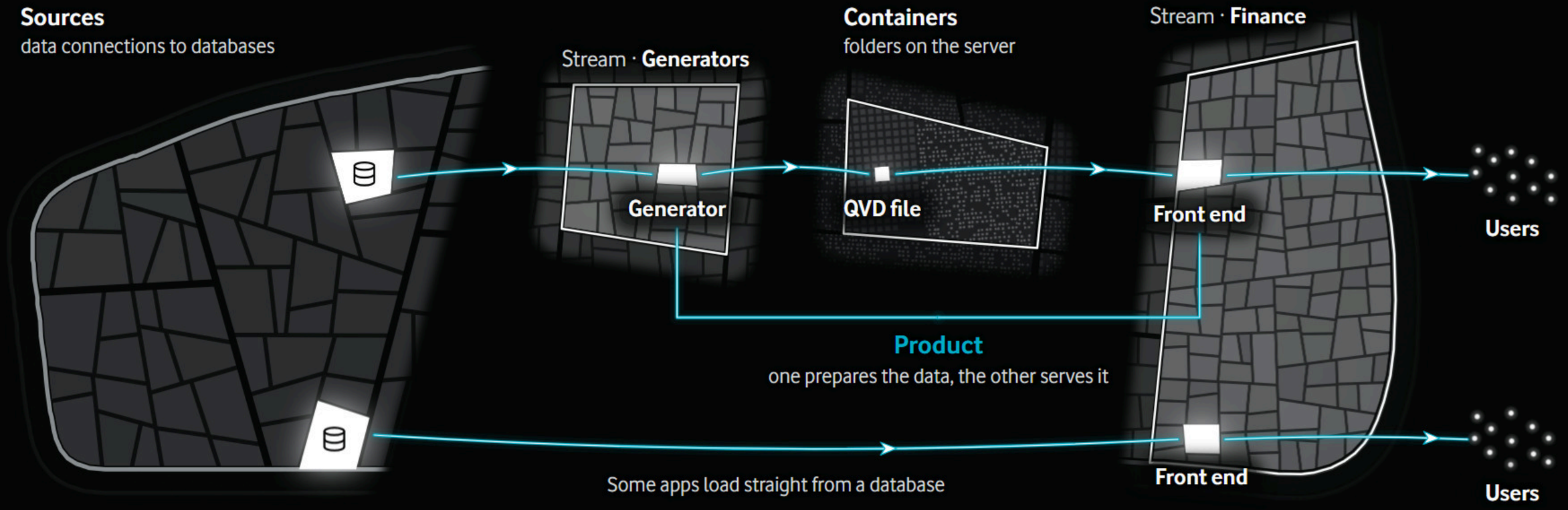
Reload the app runs its script and refreshes its data

Stream a folder in the Qlik hub, where users find their apps

Container each project's folder on the server

QVD a table saved to disk, fast to reuse

Qlik in 90 seconds



App a dashboard + the script that loads its data

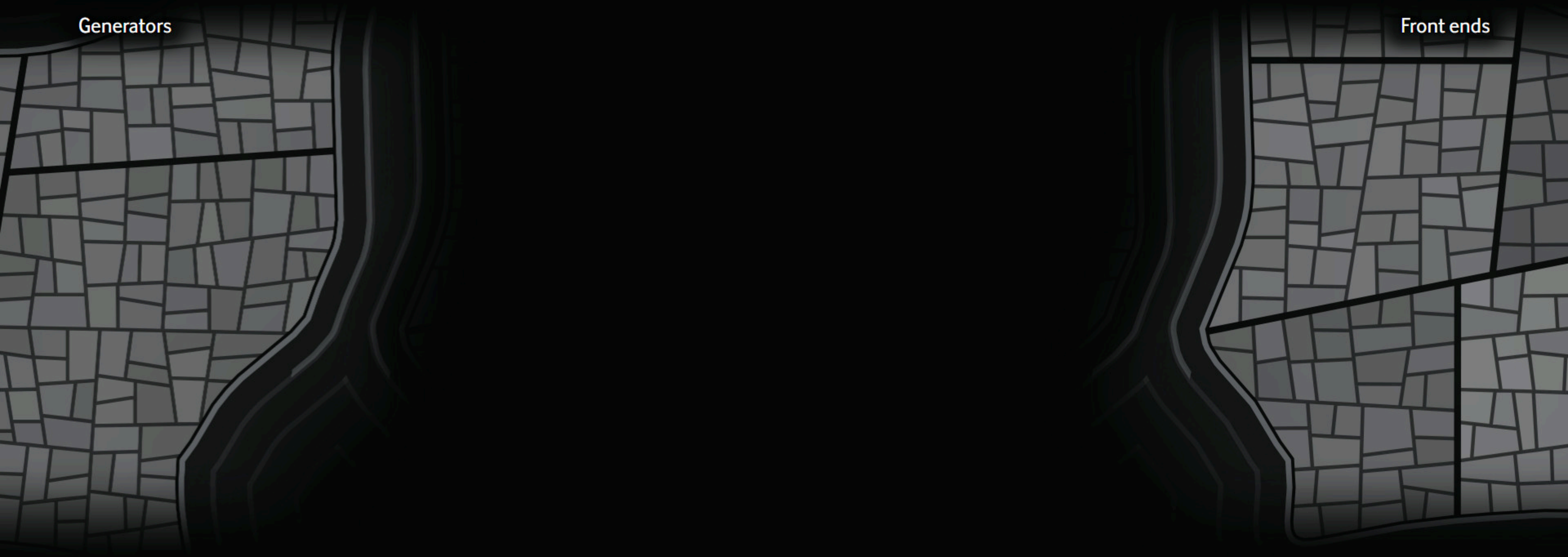
Reload the app runs its script and refreshes its data

Stream a folder in the Qlik hub, where users find their apps

Container each project's folder on the server

QVD a table saved to disk, fast to reuse

Which one can we delete?

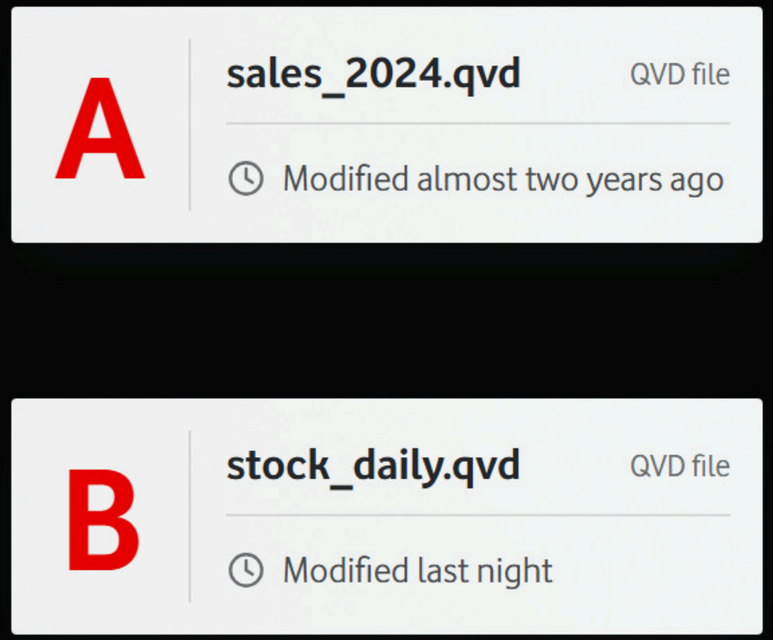


Illustrative example

Which one can we delete?

Generators

Front ends



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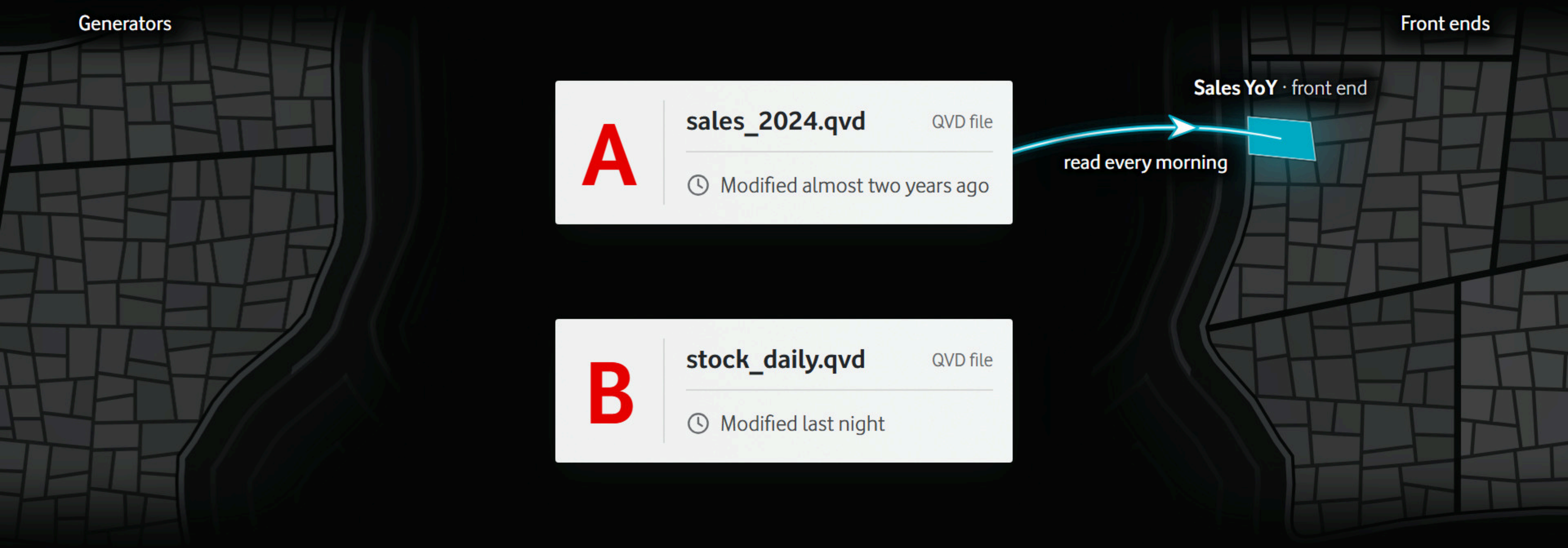

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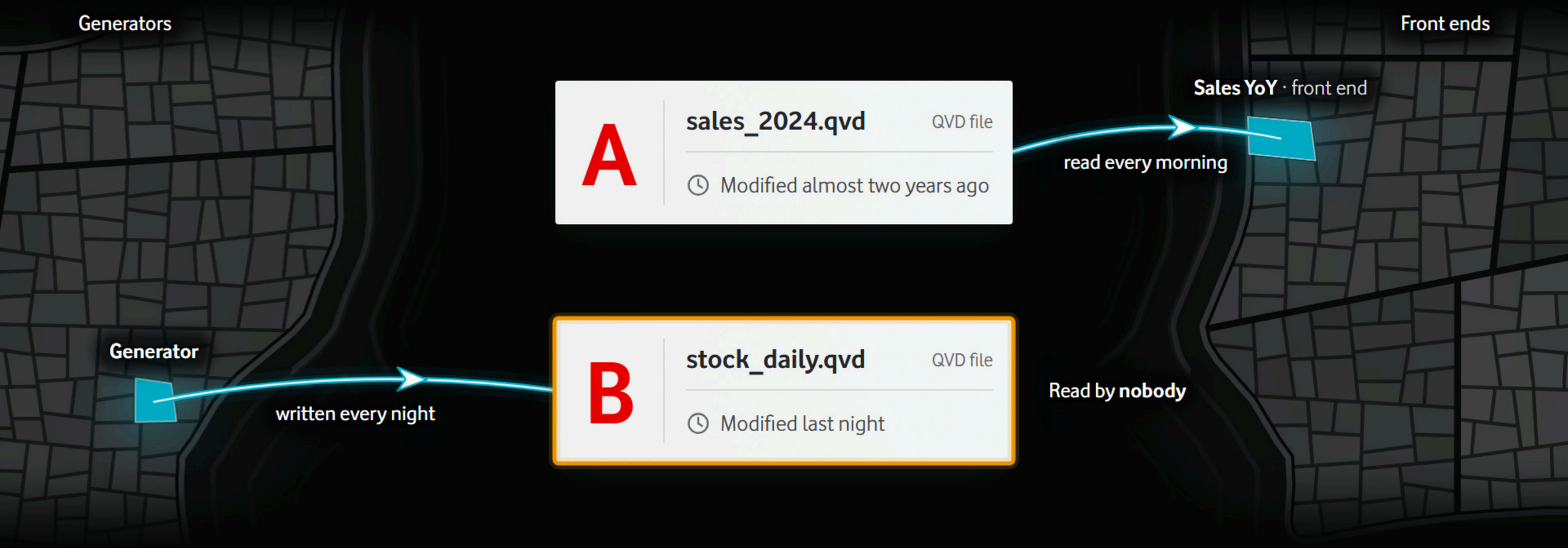

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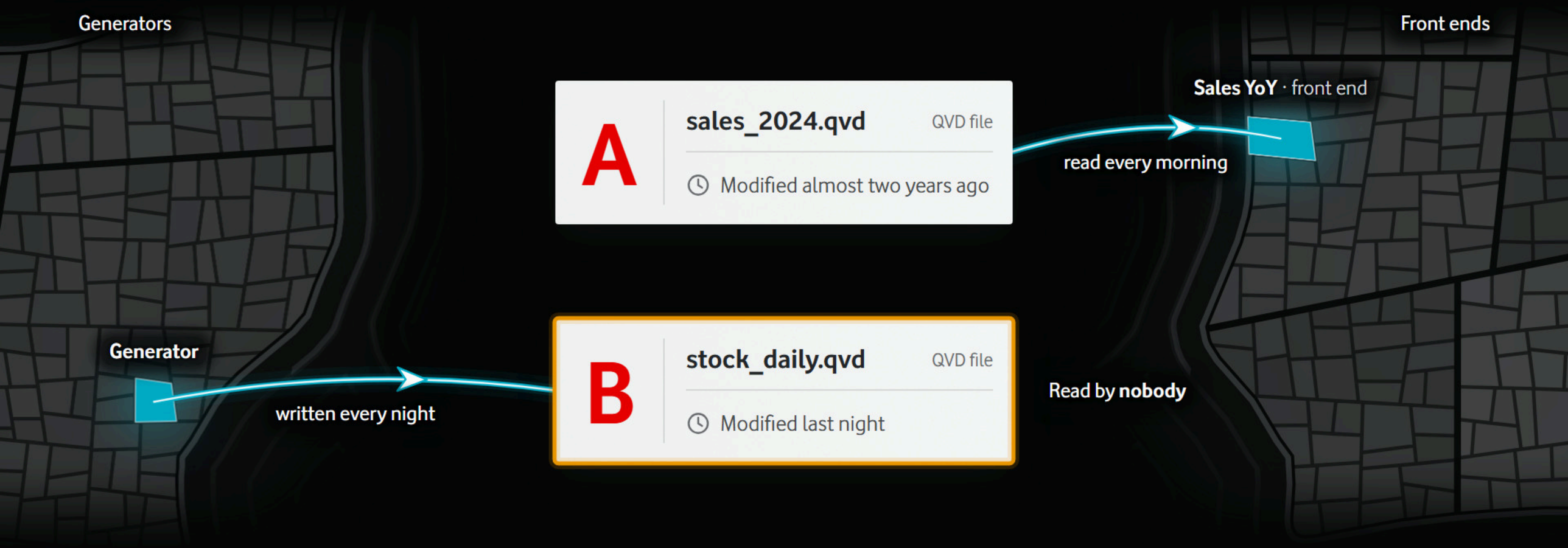


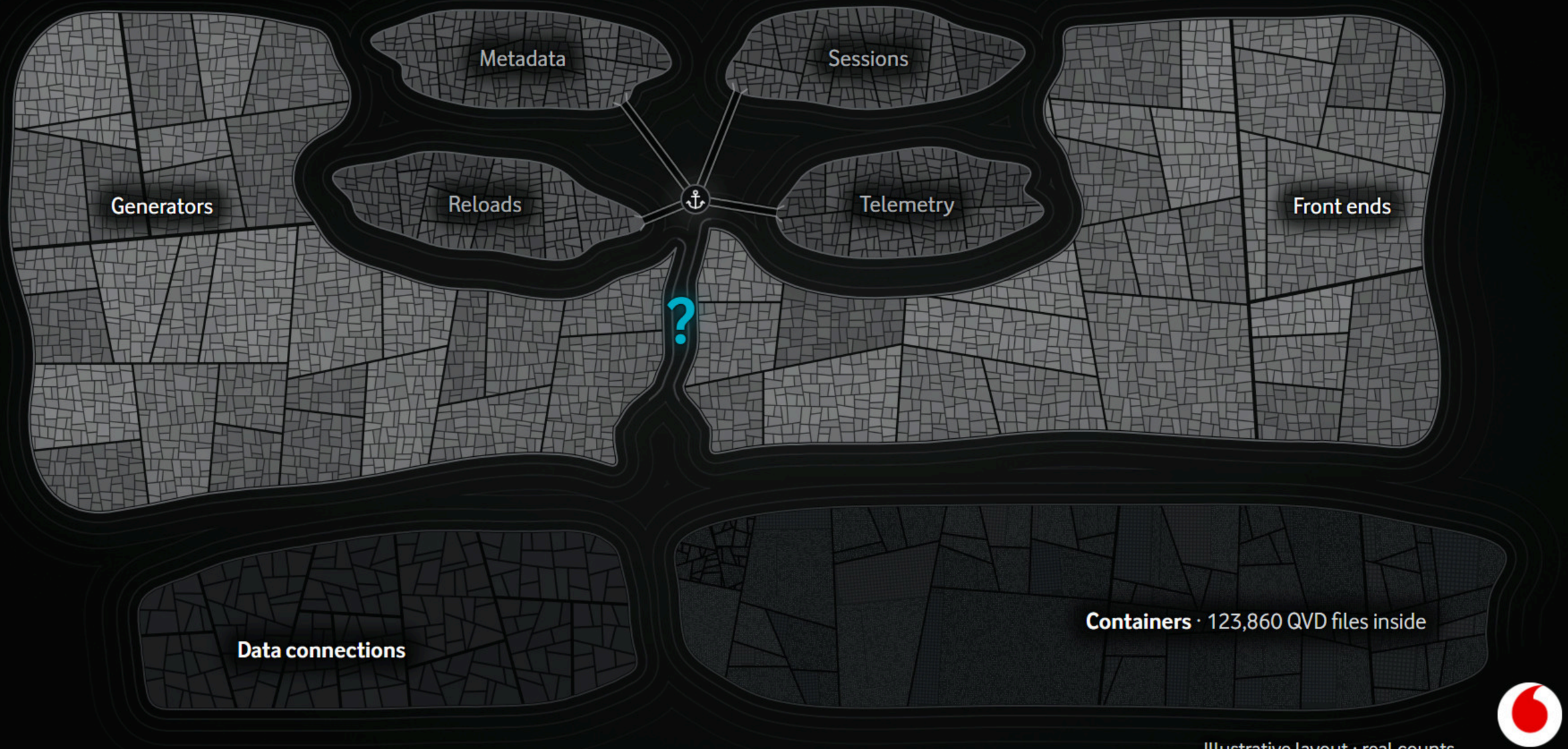
Age tells you when a file was written.
Not whether anyone still needs it.

Illustrative example

We weren't short of data

We weren't short of data

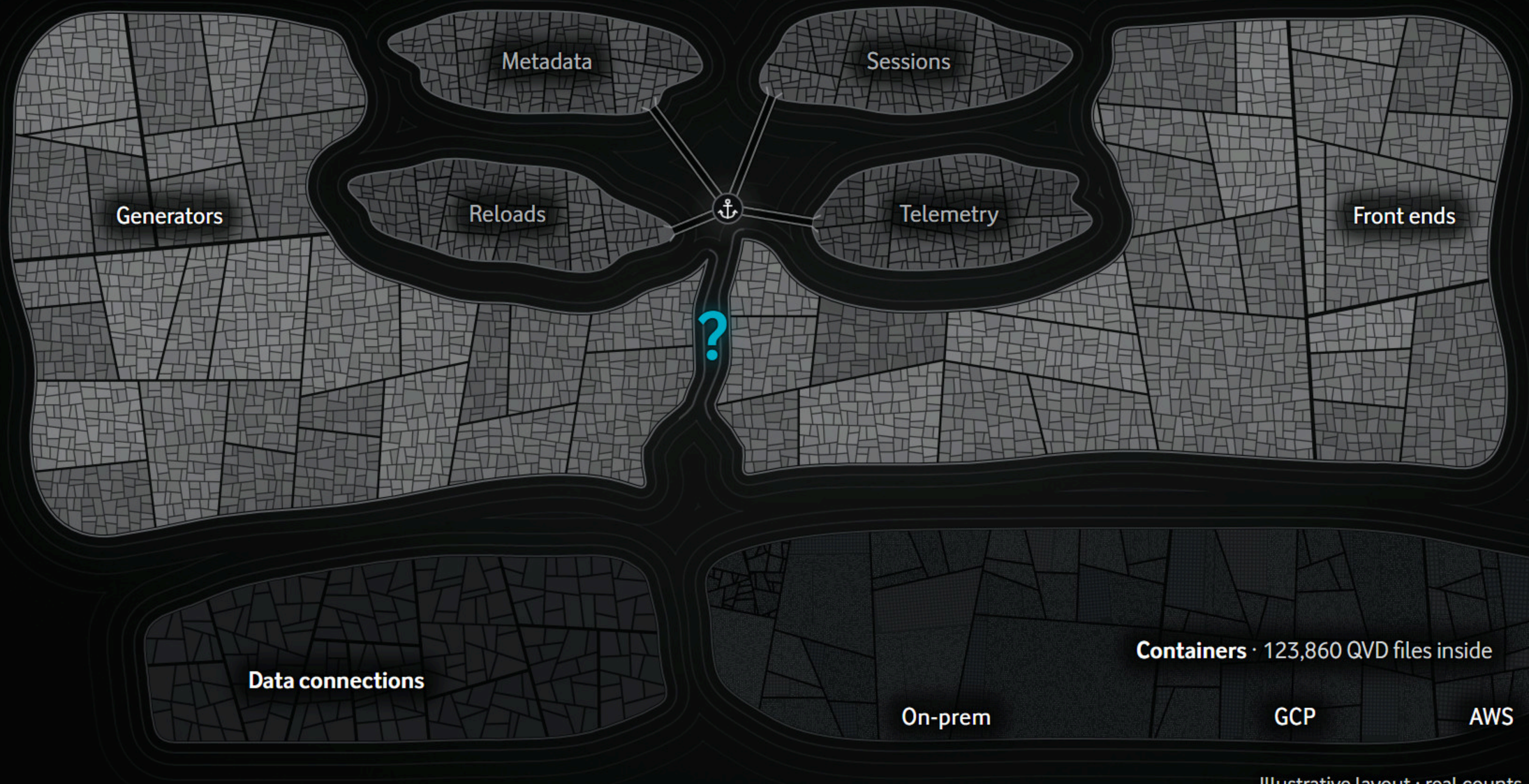
Apps · 3,562 · On-prem, GCP and AWS



Illustrative layout · real counts

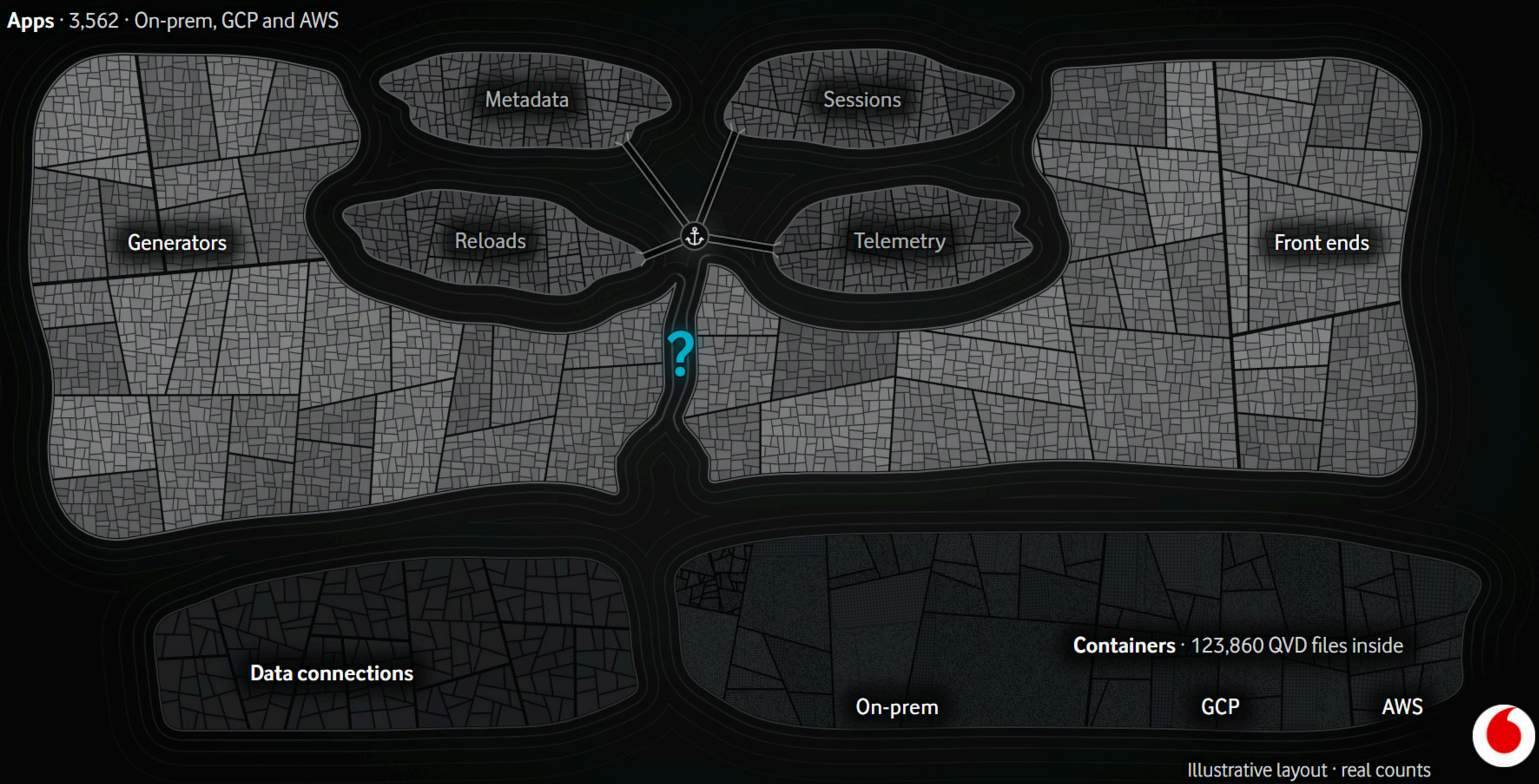
We weren't short of data

Lots of information.
Few relationships.



Illustrative layout · real counts

Four questions we couldn't answer



Illustrative layout · real counts

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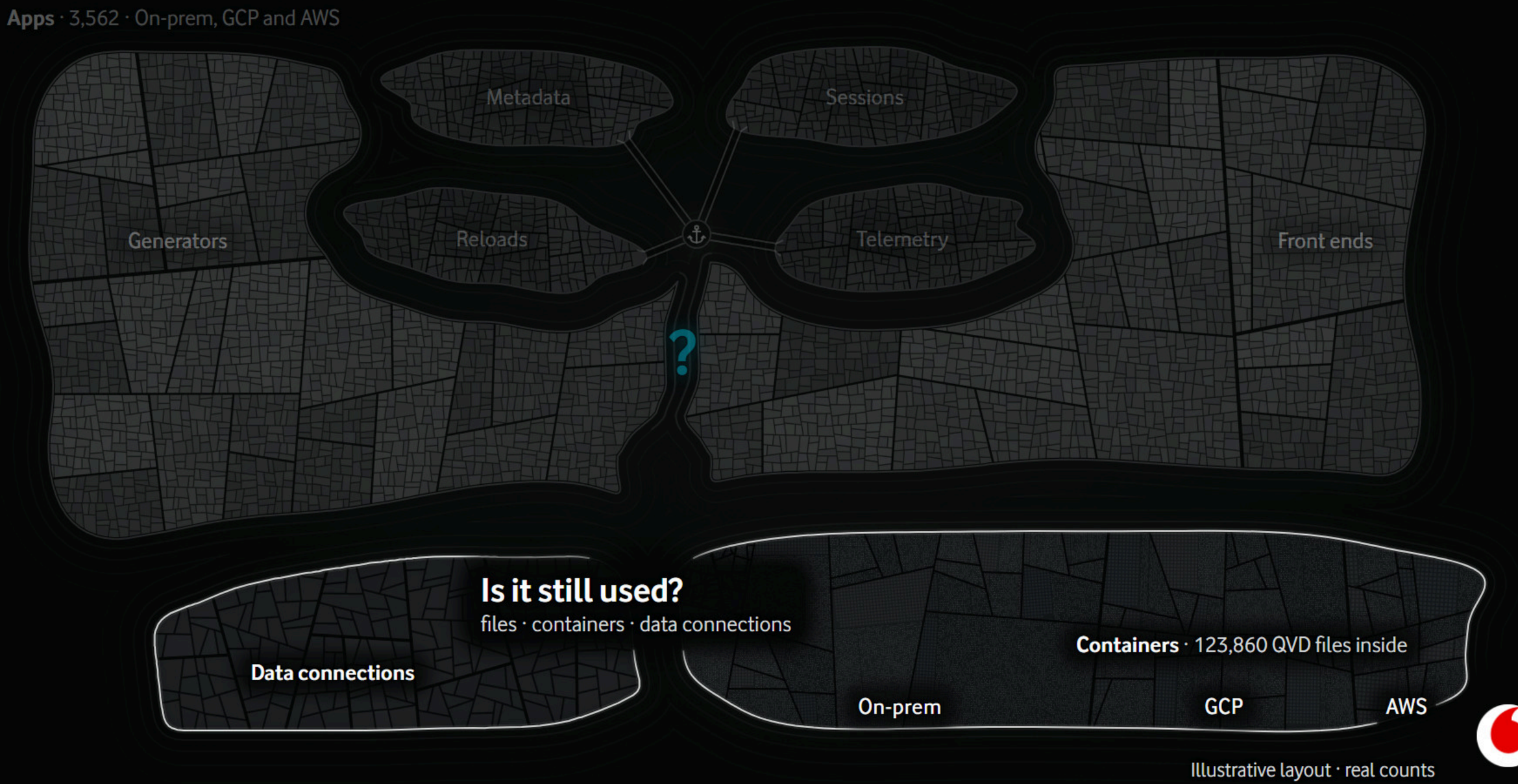

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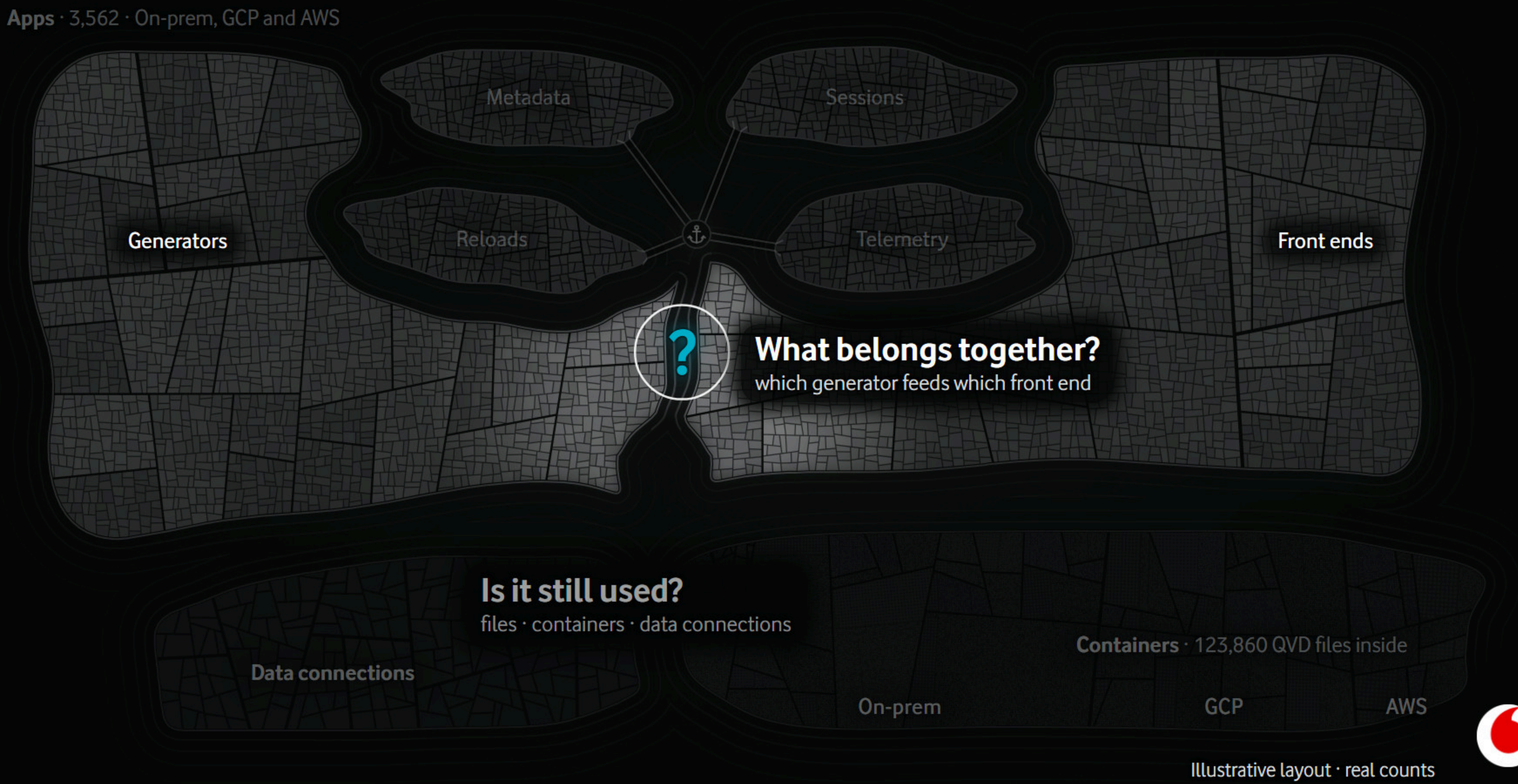

Illustrative layout · real counts

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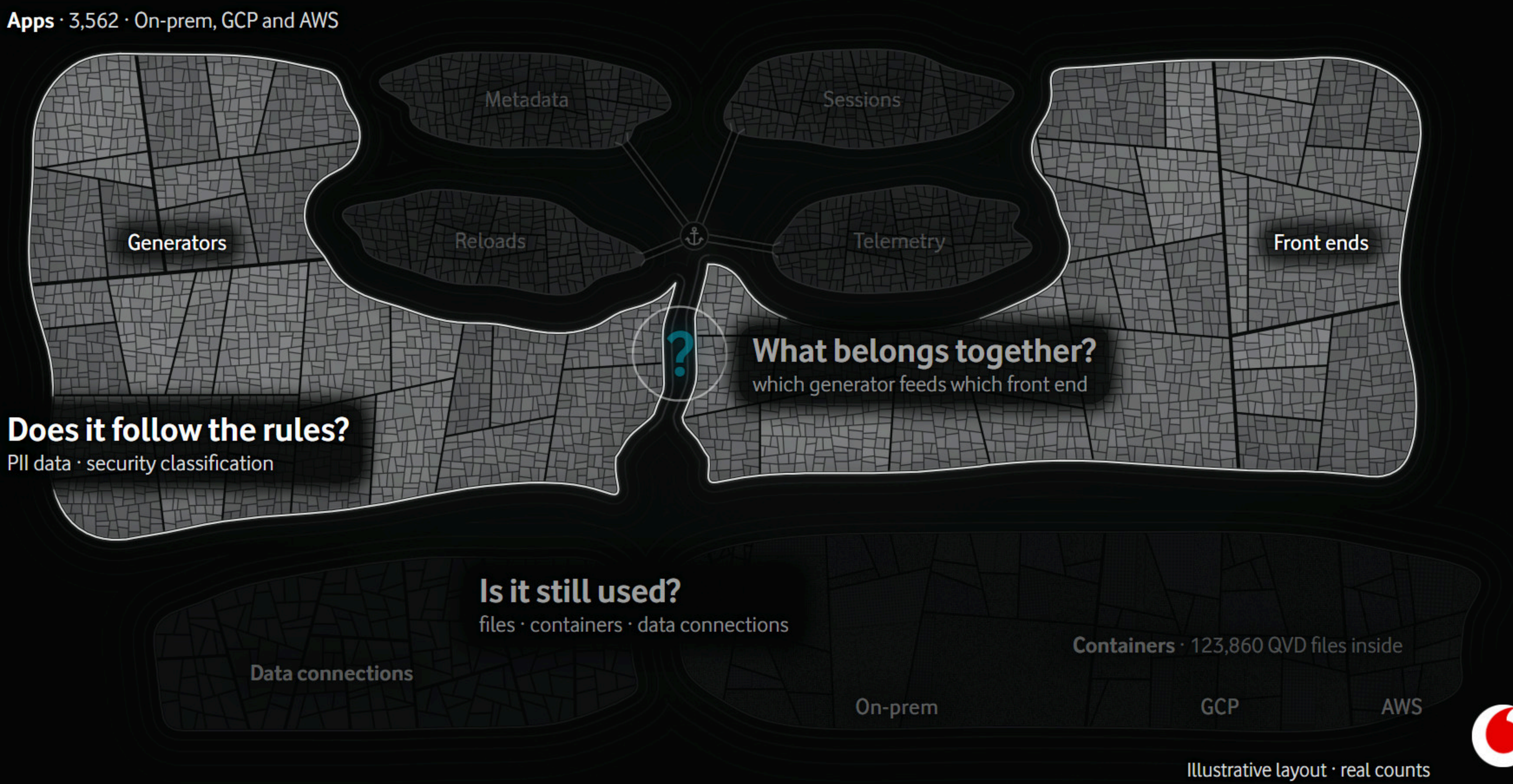
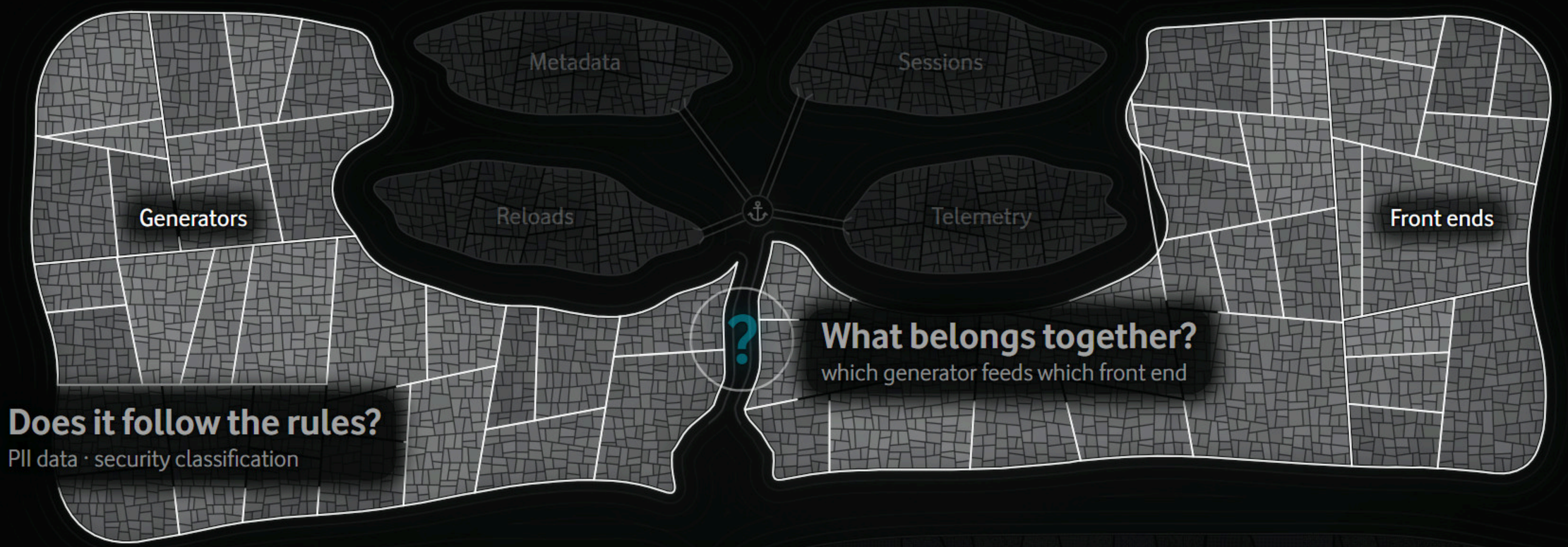


Illustrative layout · real counts

Four questions we couldn't answer

Who owns it?

Apps · 3,562 · On-prem, GCP and AWS



Is it still used?

files · containers · data connections

Data connections

Containers · 123,860 QVD files inside

On-prem GCP AWS

Illustrative layout · real counts

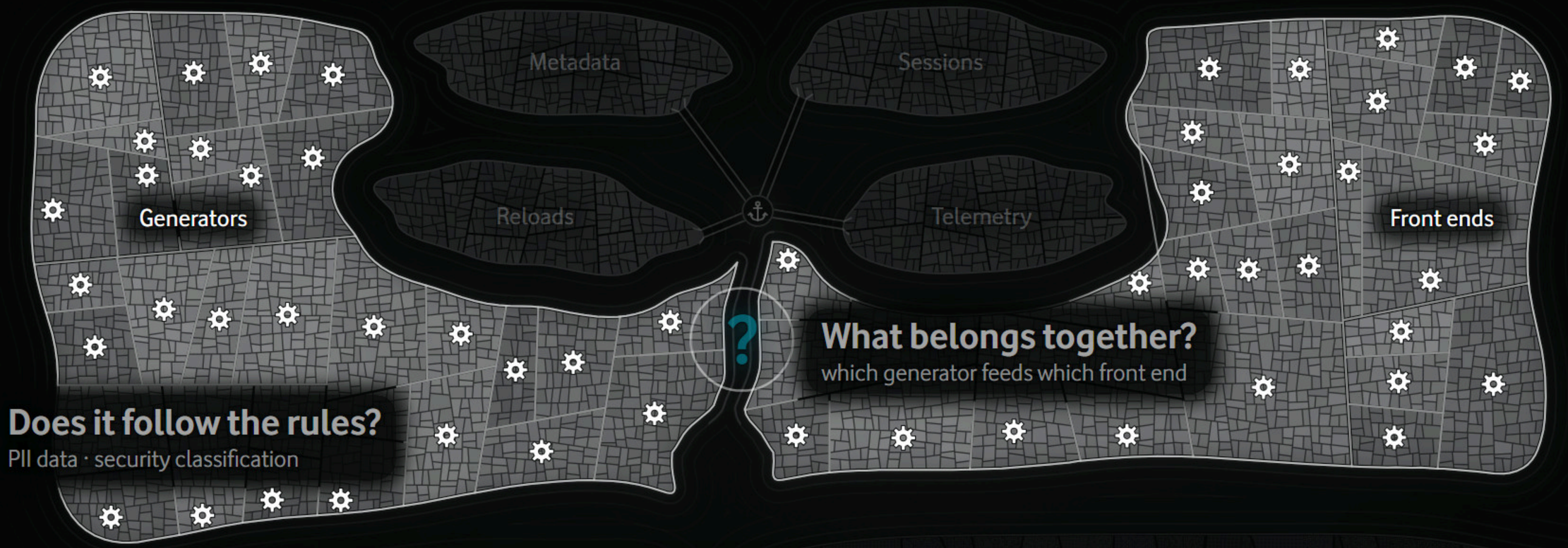
Four questions we couldn't answer

Apps · 3,562 · On-prem, GCP and AWS

Who owns it?

Owner: ⚙ a technical account

Ops publishes to Production



What belongs together?

which generator feeds which front end

Does it follow the rules?

PII data · security classification

Is it still used?

files · containers · data connections

Containers · 123,860 QVD files inside

Data connections

On-prem

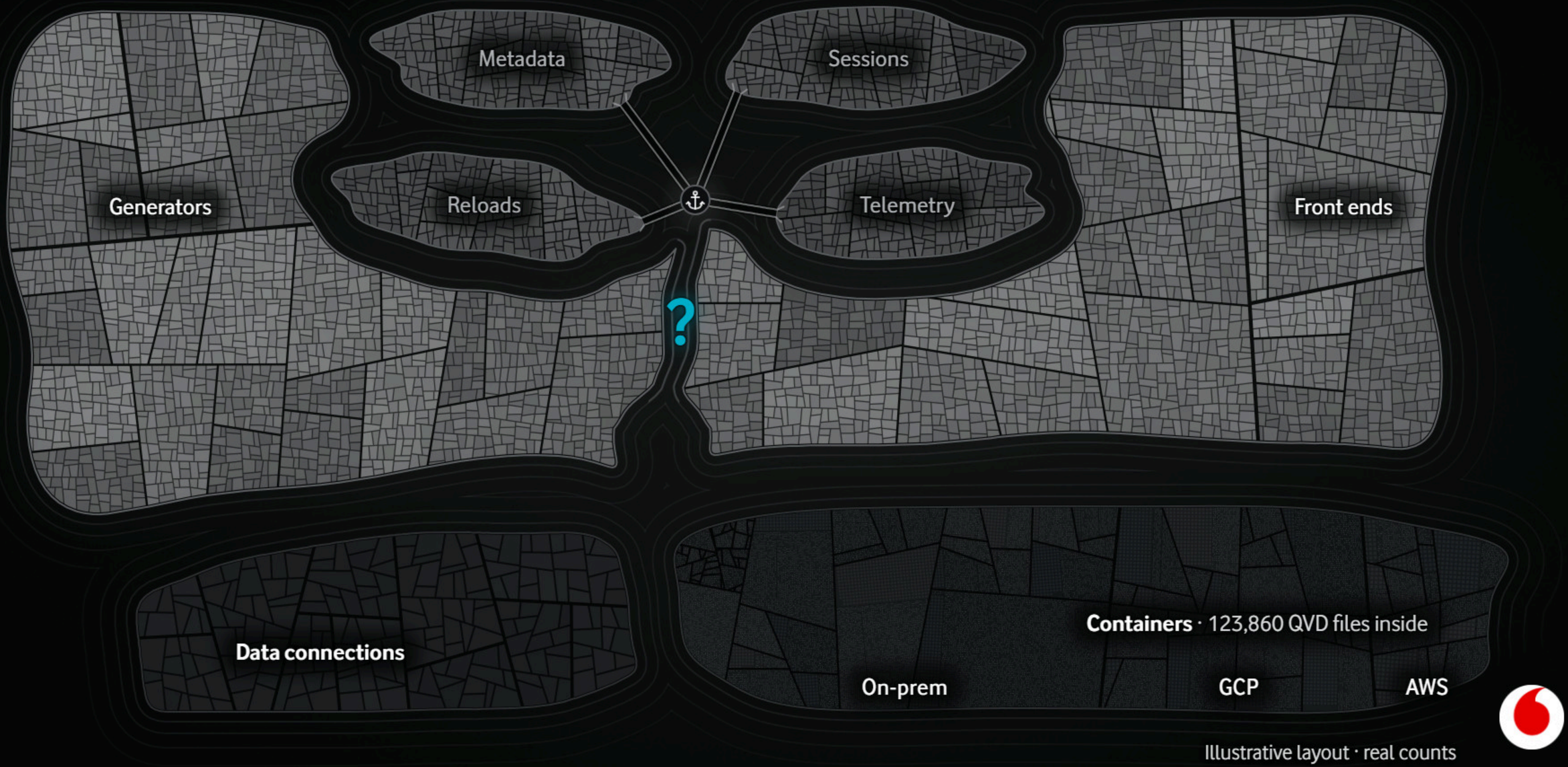
GCP

AWS

Illustrative layout · real counts

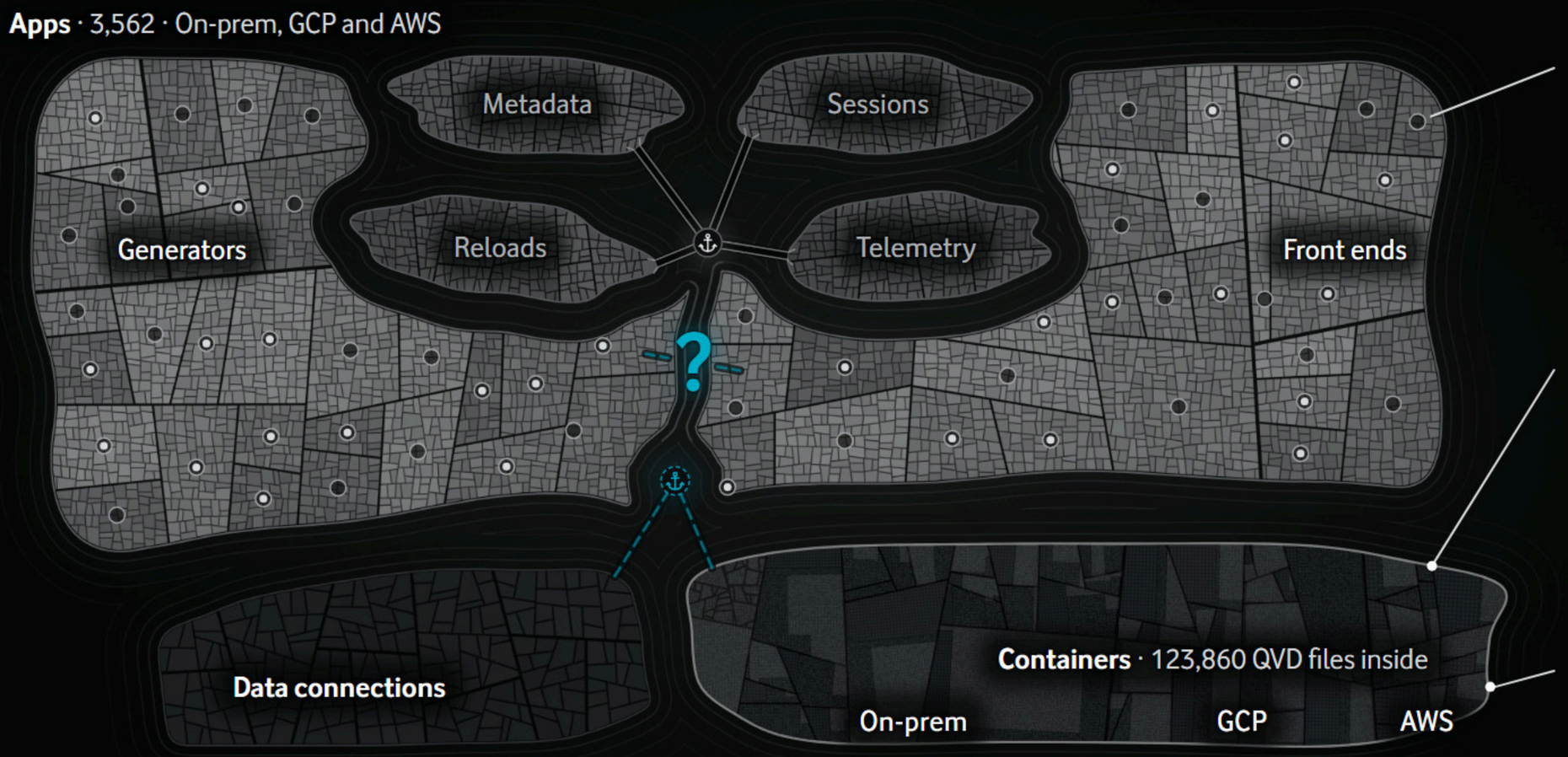
The map we need

Apps · 3,562 · On-prem, GCP and AWS



Illustrative layout · real counts

The map we need



Owners

can't see Production folders:
Ops moves the data

PII data

old files may keep it
past retention

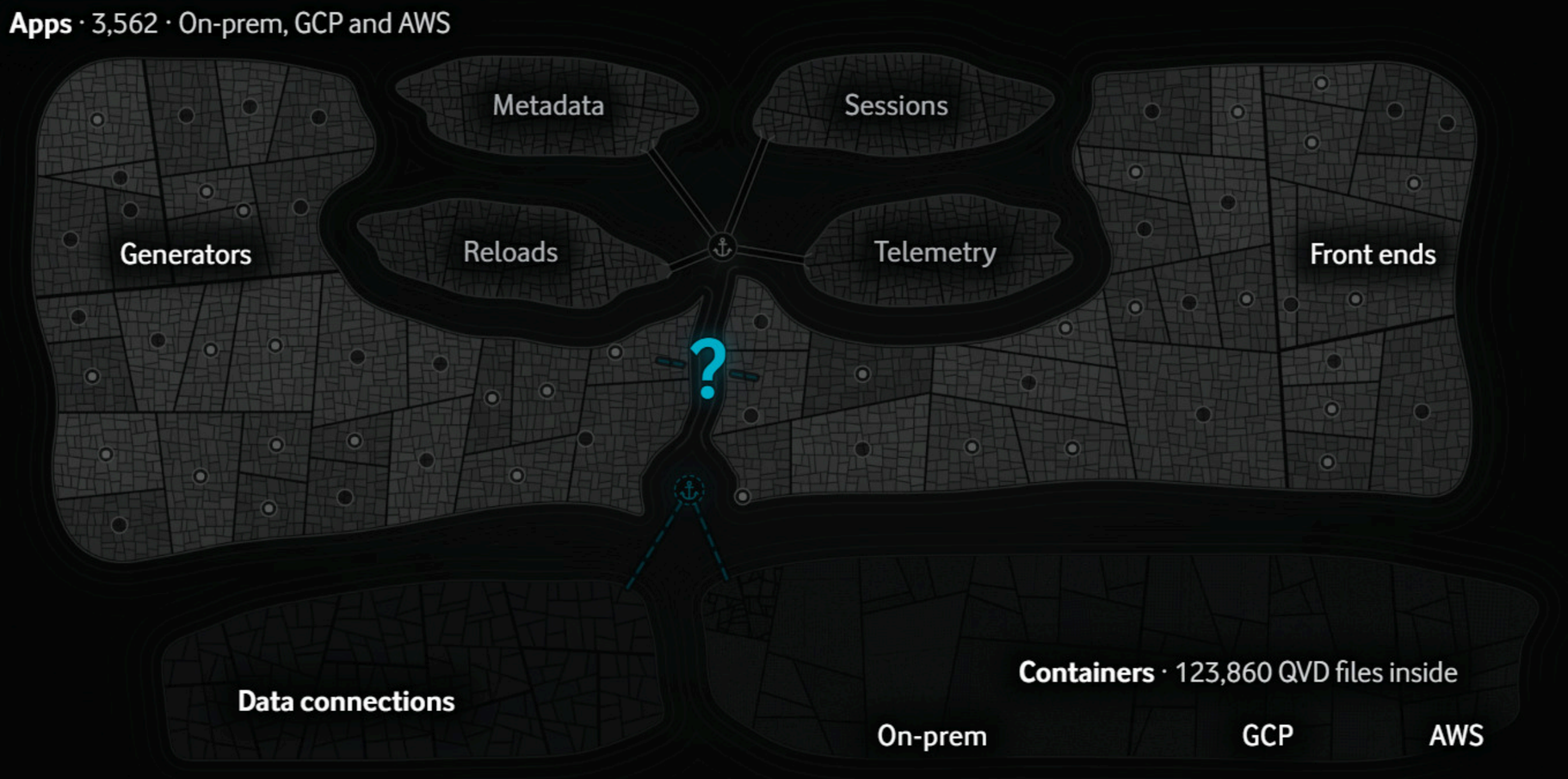
Space

usually less than 100 GB free

what we have · what we need · an owner per stream · Ops, in progress

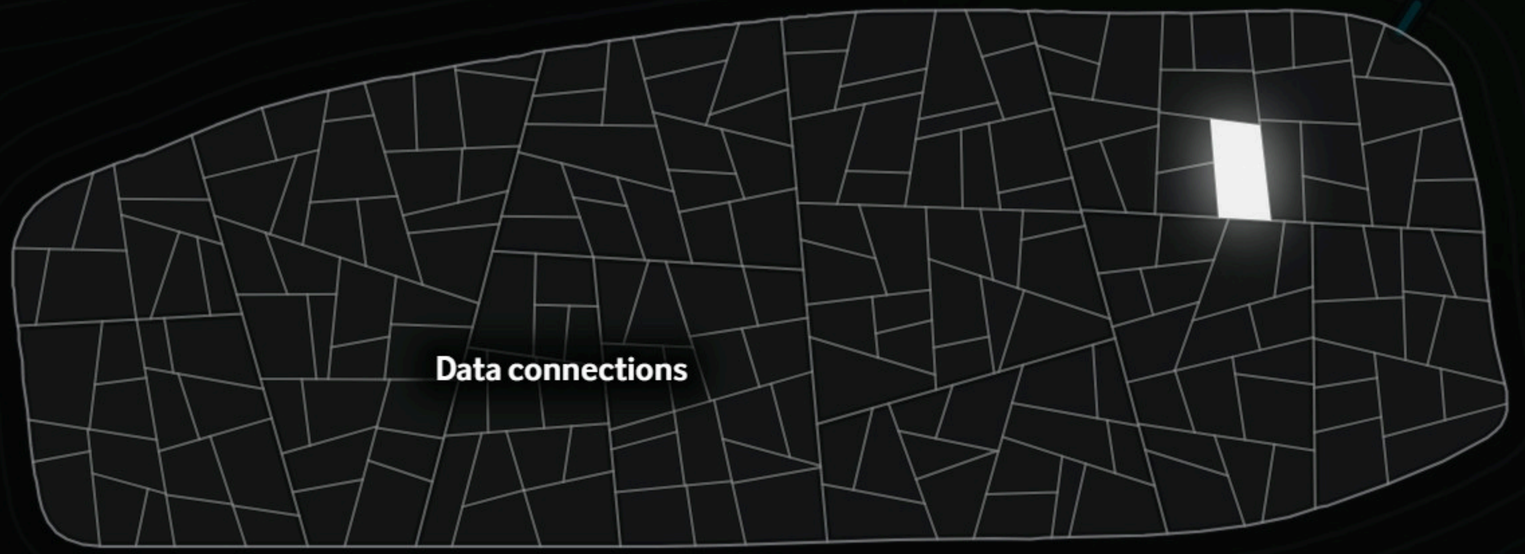
Illustrative layout · real counts

Then came a smaller question



Illustrative layout · real counts

Which data connections are actually used?



Illustrative layout · real counts

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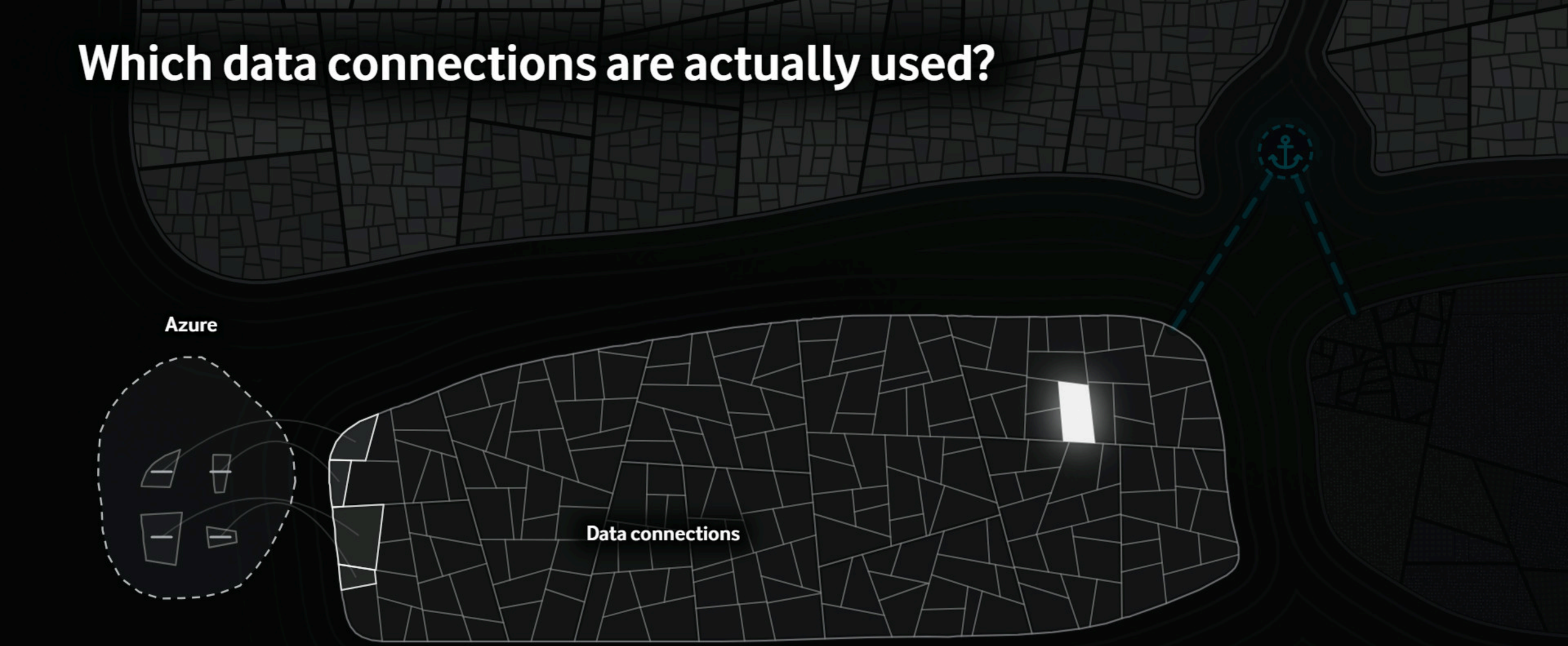


Migrating an unused connection
= effort with no value

Illustrative layout · real counts

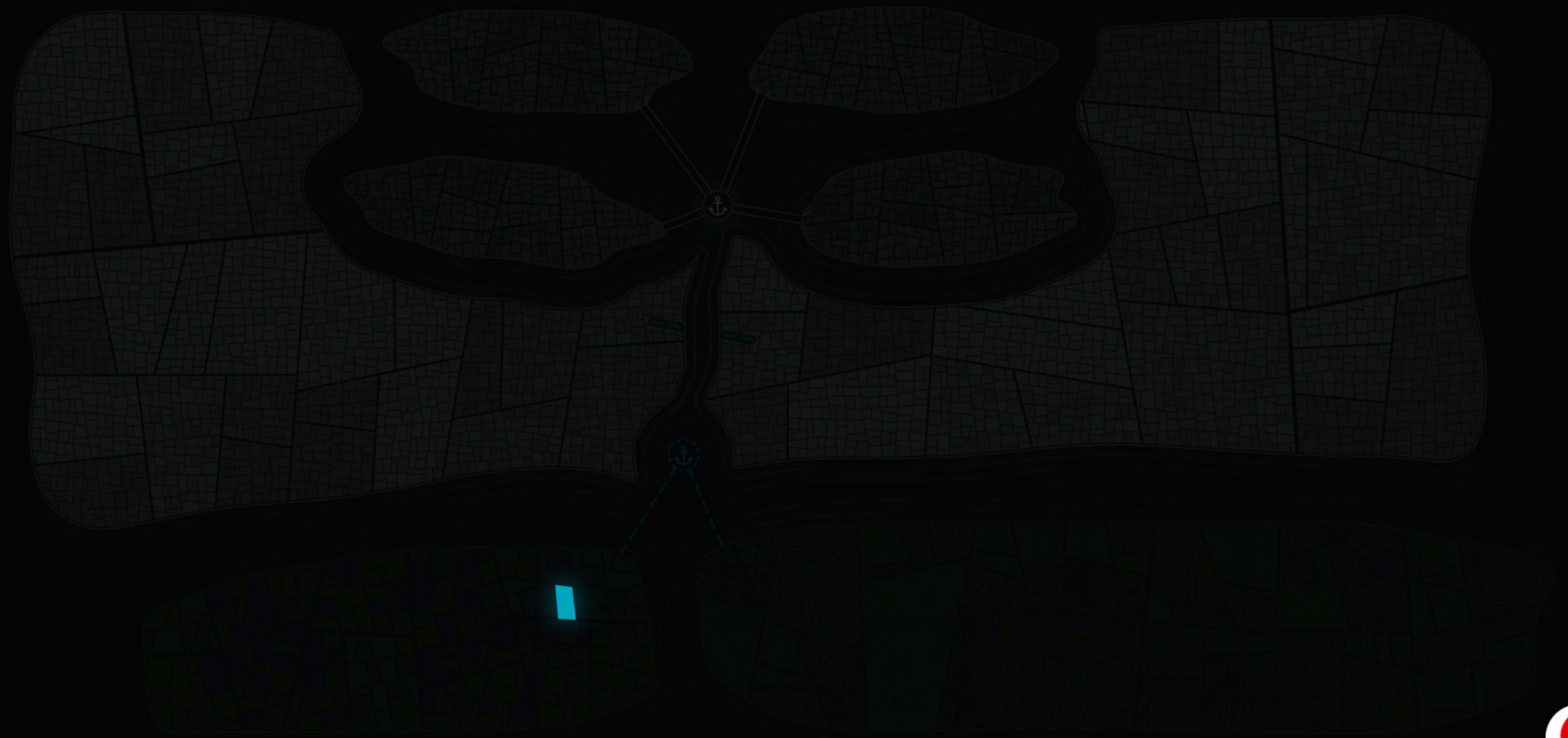
Which data connections are actually used?

Illustrative layout · real counts

How did it know?

Illustrative layout · real counts

The answer was in the diary



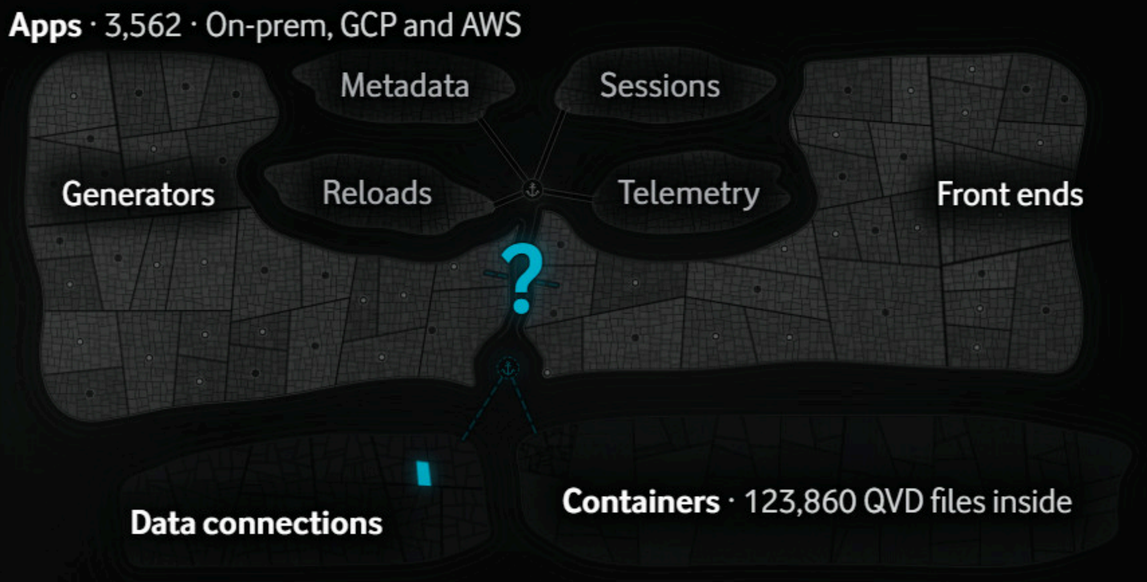
Illustrative layout · real counts

The answer was in the diary

Generator app — reload log · 12 Oct 2026 · 06:00

```
Execution started.
Qlik Sense Server Version 14.212.6
0002  SET ThousandSep=','
0004  SET DateFormat='DD/MM/YYYY'
0012  // ---- Template: mandatory governance variables ----
0013  LET vPII = 'Yes'
0014  LET vSecurityClass = 'Confidential'
0021  // ---- Extract: sales orders ----
0022  LIB CONNECT TO 'SALES_DB'
Connected.
0024  SQL SELECT * FROM dbo.SalesOrders
14 fields found: OrderID, OrderDate, CustomerID, ProductID, Quantity, NetAmount, ...
1,284,519 Lines fetched
0026  STORE SalesOrders INTO [lib://SalesContainer/QVD/sales_orders.qvd] (qvd)
0027  DROP TABLE SalesOrders
0031  // ---- Extract: customers ----
0032  Customers:
0033  SQL SELECT CustomerID, Segment, Region FROM dbo.Customers
3 fields found: CustomerID, Segment, Region
86,204 Lines fetched
0035  STORE Customers INTO [lib://SalesContainer/QVD/customers.qvd] (qvd)
0036  DROP TABLE Customers
0041  DISCONNECT
Execution finished.
```

Every reload writes a diary



Illustrative layout · real counts

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Generator app — reload log · 12 Oct 2026 · 06:00

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        Execution started.
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0022    LIB CONNECT TO 'SALES_DB'
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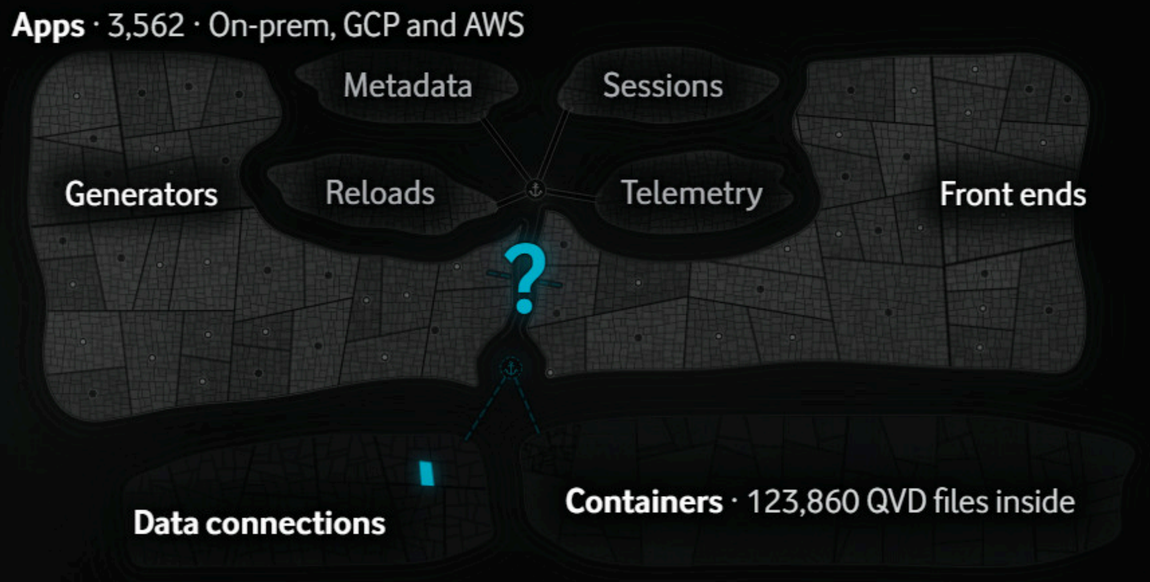
```

LIB CONNECT TO 'SALES_DB'



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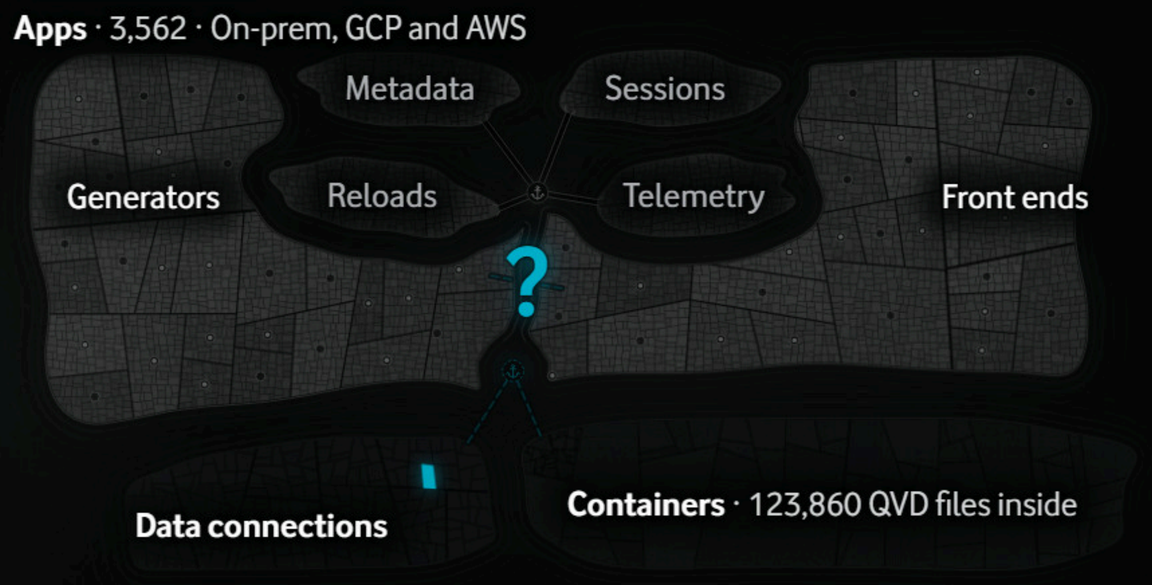
LIB CONNECT TO 'SALES_DB'

— connected to a database



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0041   DISCONNECT
```

LIB CONNECT TO 'SALES_DB'

— connected to a database

Illustrative layout · real counts

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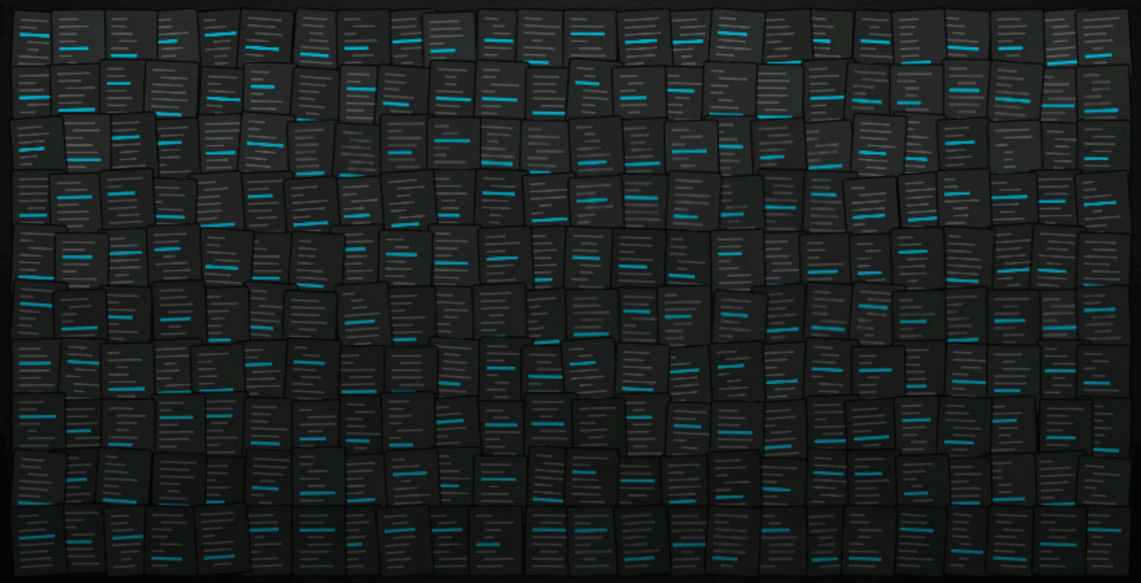

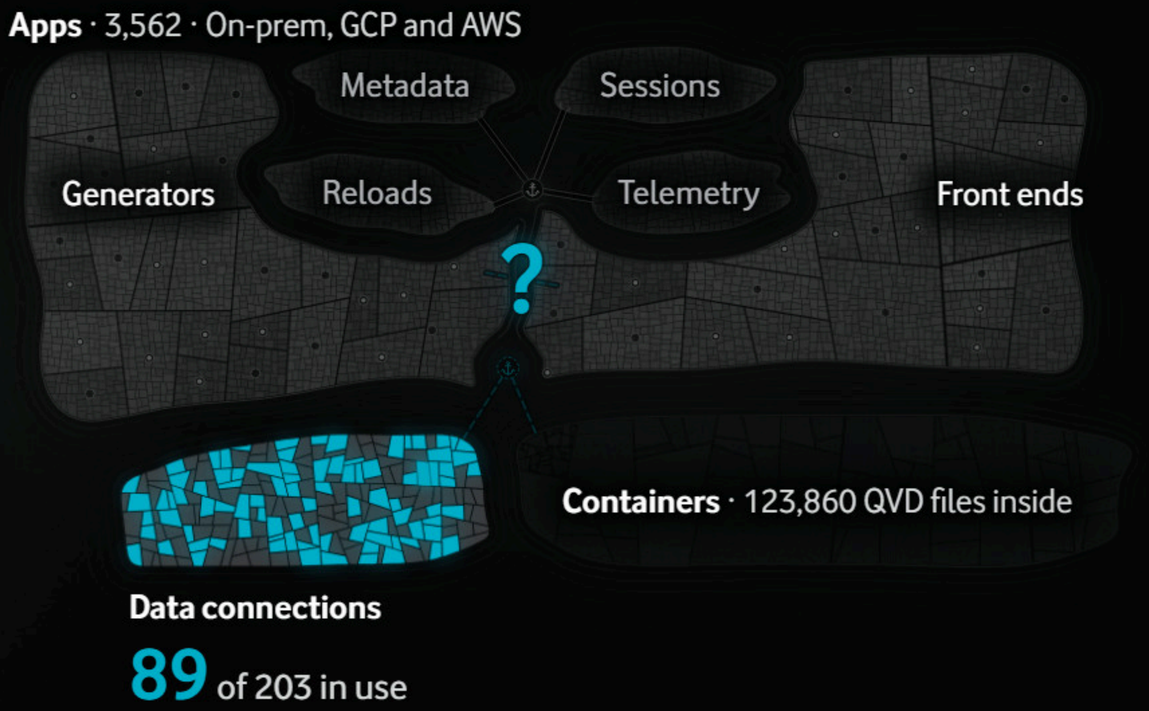

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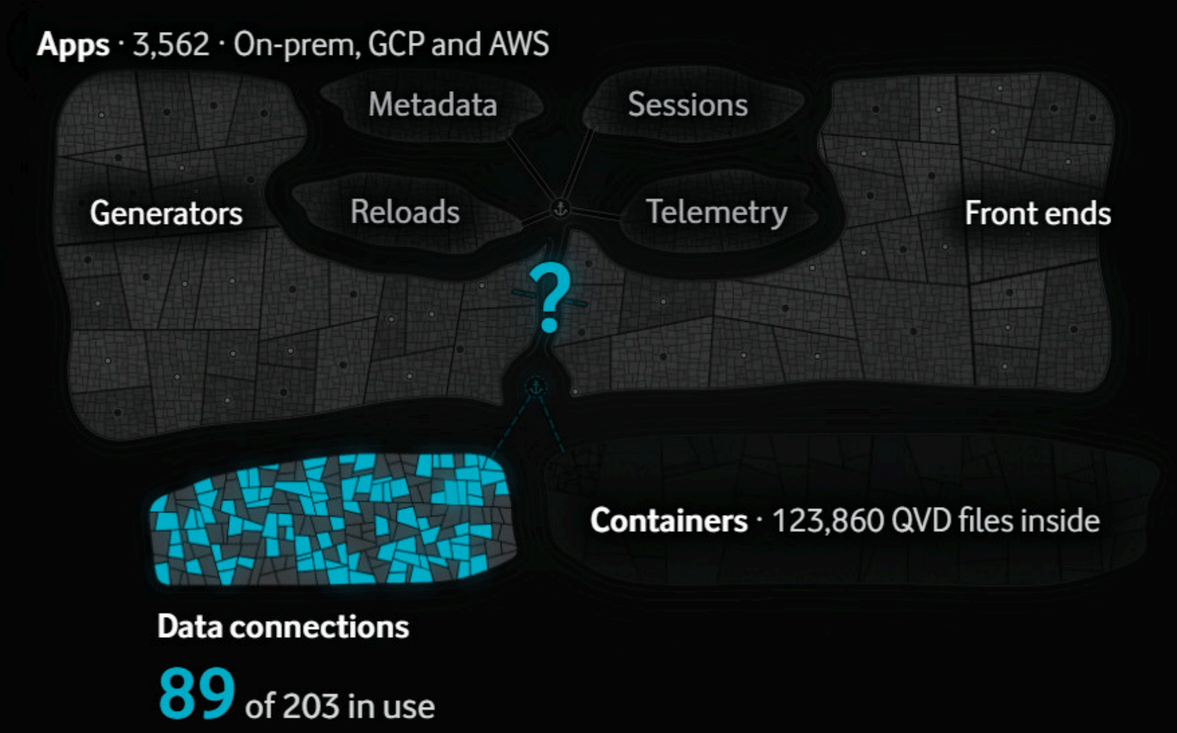

It didn't find the list.
It built it.

Illustrative layout · real counts

One clue became a method

Generator app — reload log · 12 Oct 2026 · 06:00

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        Execution started.
0013    LET vPII = 'Yes'
0021    // ---- Extract: sales orders ----
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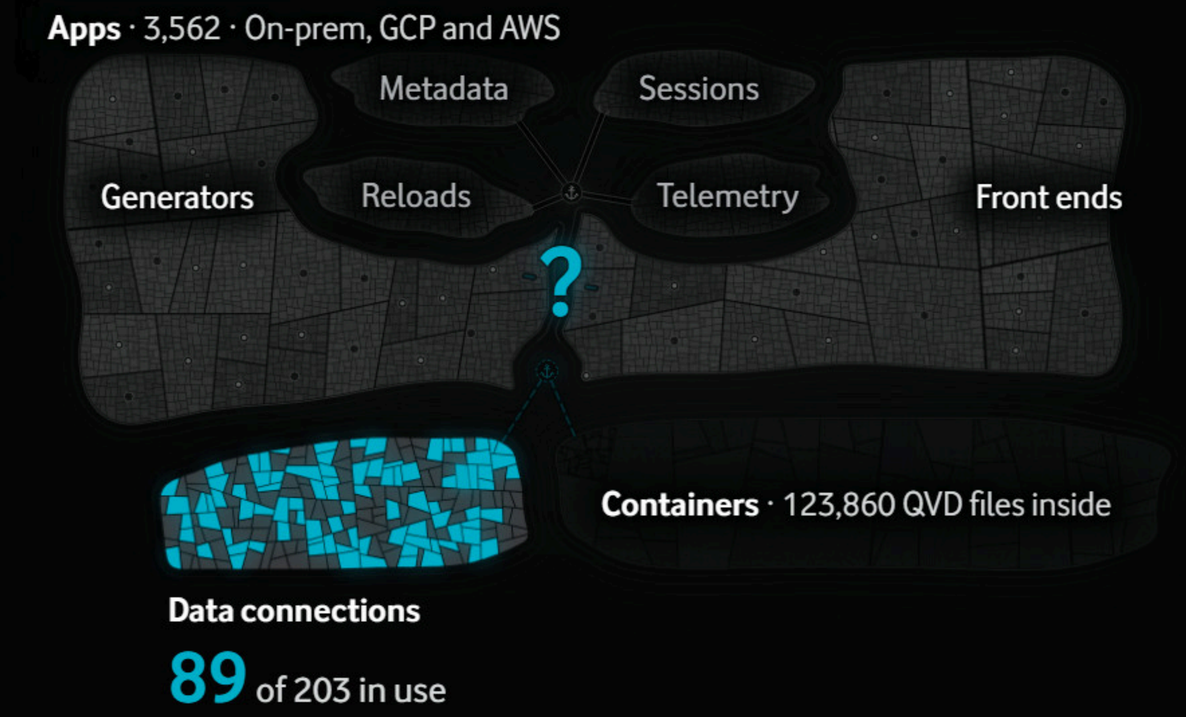
        — connected to a database   the clue we just found
0024    SQL SELECT * FROM dbo.SalesOrders
0026    STORE SalesOrders INTO [lib://SalesContainer/QVD/sales_orders.qvd] (qvd)
0027    DROP TABLE SalesOrders
0041    DISCONNECT
```



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        — saved a file
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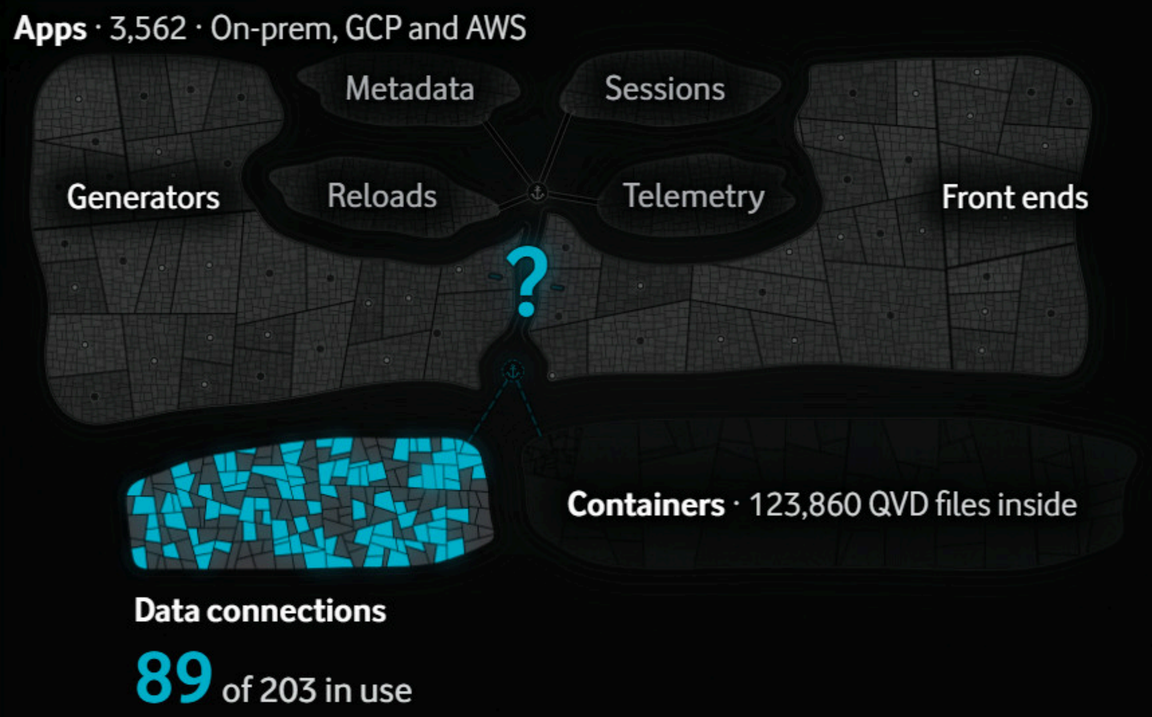

Illustrative layout · real counts

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        — saved a file
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0041    DISCONNECT
```
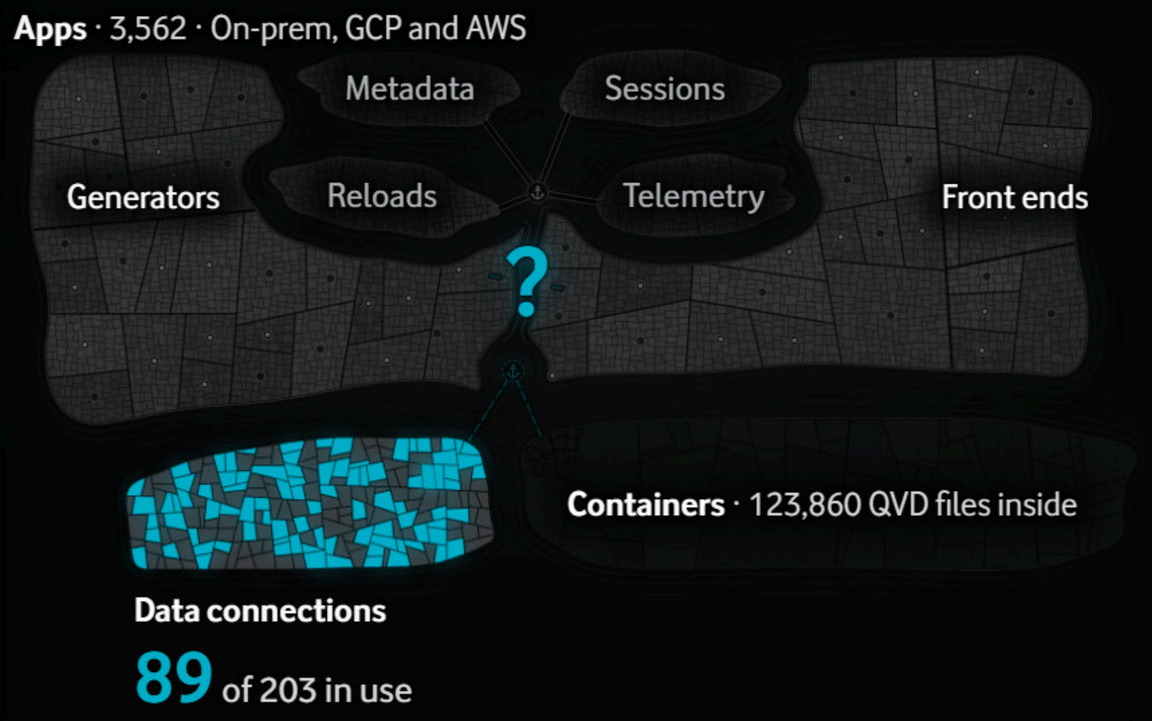

Sales YoY front end reload log · 12 Oct 2026 · 07:02

```
        Execution started.
0011    LET vPII = 'Yes'
0018    // ---- Data model ----
0020    LOAD * FROM [lib://SalesContainer/QVD/sales_orders.qvd] (qvd)
        — read a file
0024    LOAD * FROM [lib://SalesContainer/QVD/customers.qvd] (qvd)
0028    LOAD * FROM [lib://MasterContainer/QVD/calendar.qvd] (qvd)
```



Illustrative layout · real counts

One clue became a method

Generator app reload log · 12 Oct 2026 · 06:00

```
        Execution started.
0013    LET vPII = 'Yes'
        — declared: holds PII data
0021    // ---- Extract: sales orders ----
0022    LIB CONNECT TO 'SALES_DB'
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0024    SQL SELECT * FROM dbo.SalesOrders
0026    STORE SalesOrders INTO [lib://SalesContainer/QVD/sales_orders.qvd] (qvd)
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0028    LOAD * FROM [lib://MasterContainer/QVD/calendar.qvd] (qvd)
```
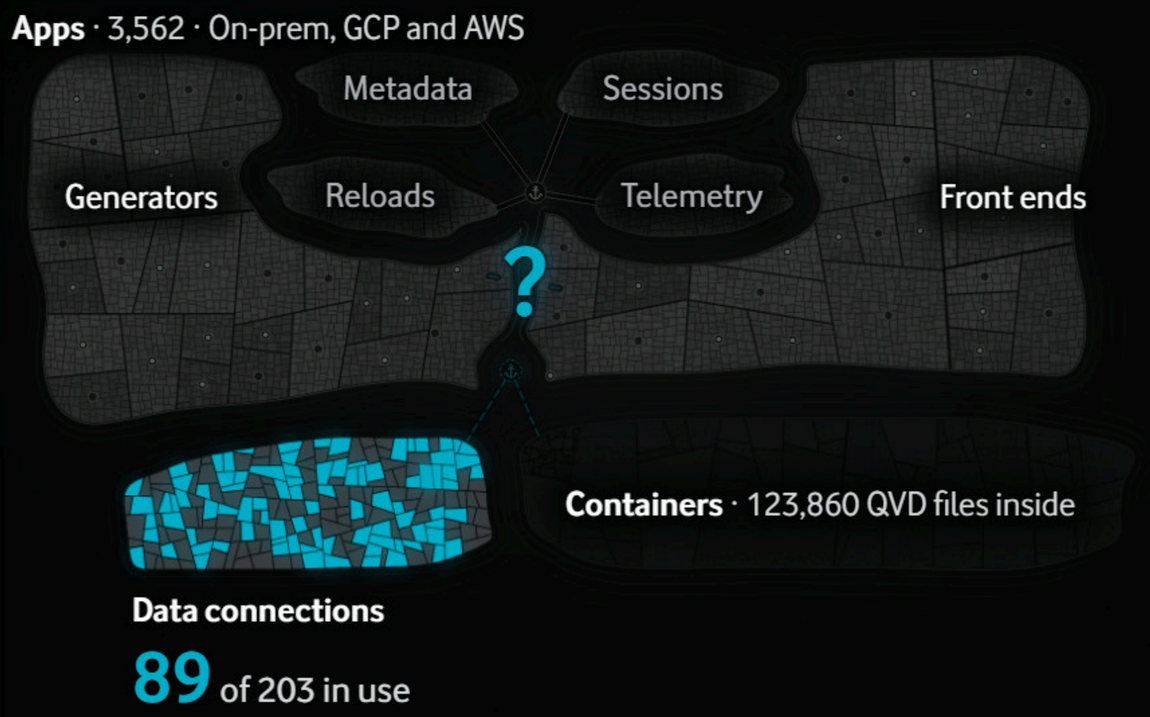



Illustrative layout · real counts

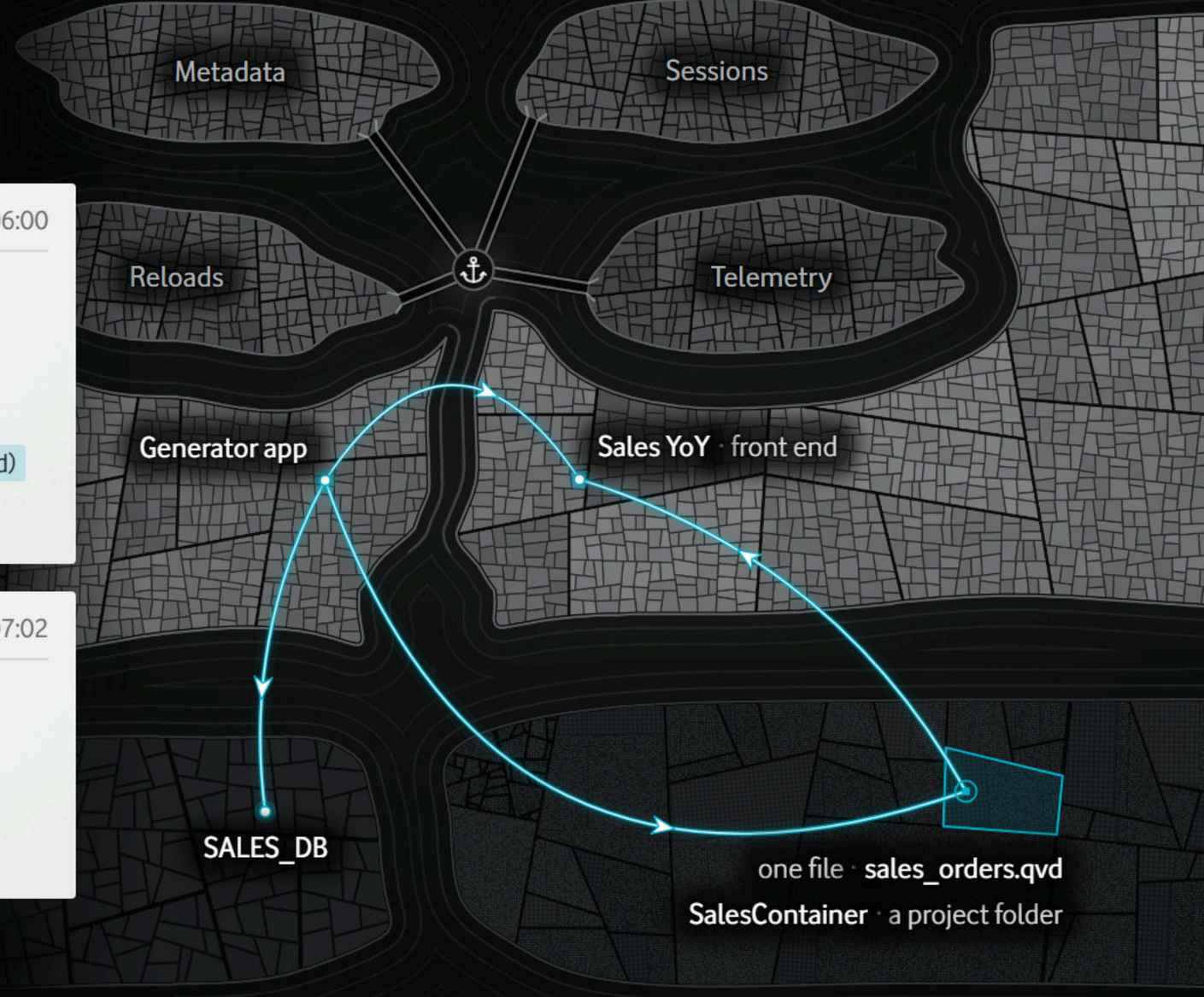
One clue became a method

Generator app reload log · 12 Oct 2026 · 06:00

```
        Execution started.
0013    LET vPII = 'Yes'
0021    // ---- Extract: sales orders ----
0022    LIB CONNECT TO 'SALES_DB'
0024    SQL SELECT * FROM dbo.SalesOrders
0026    STORE SalesOrders INTO [lib://SalesContainer/QVD/sales_orders.qvd] (qvd)
0027    DROP TABLE SalesOrders
0041    DISCONNECT
```

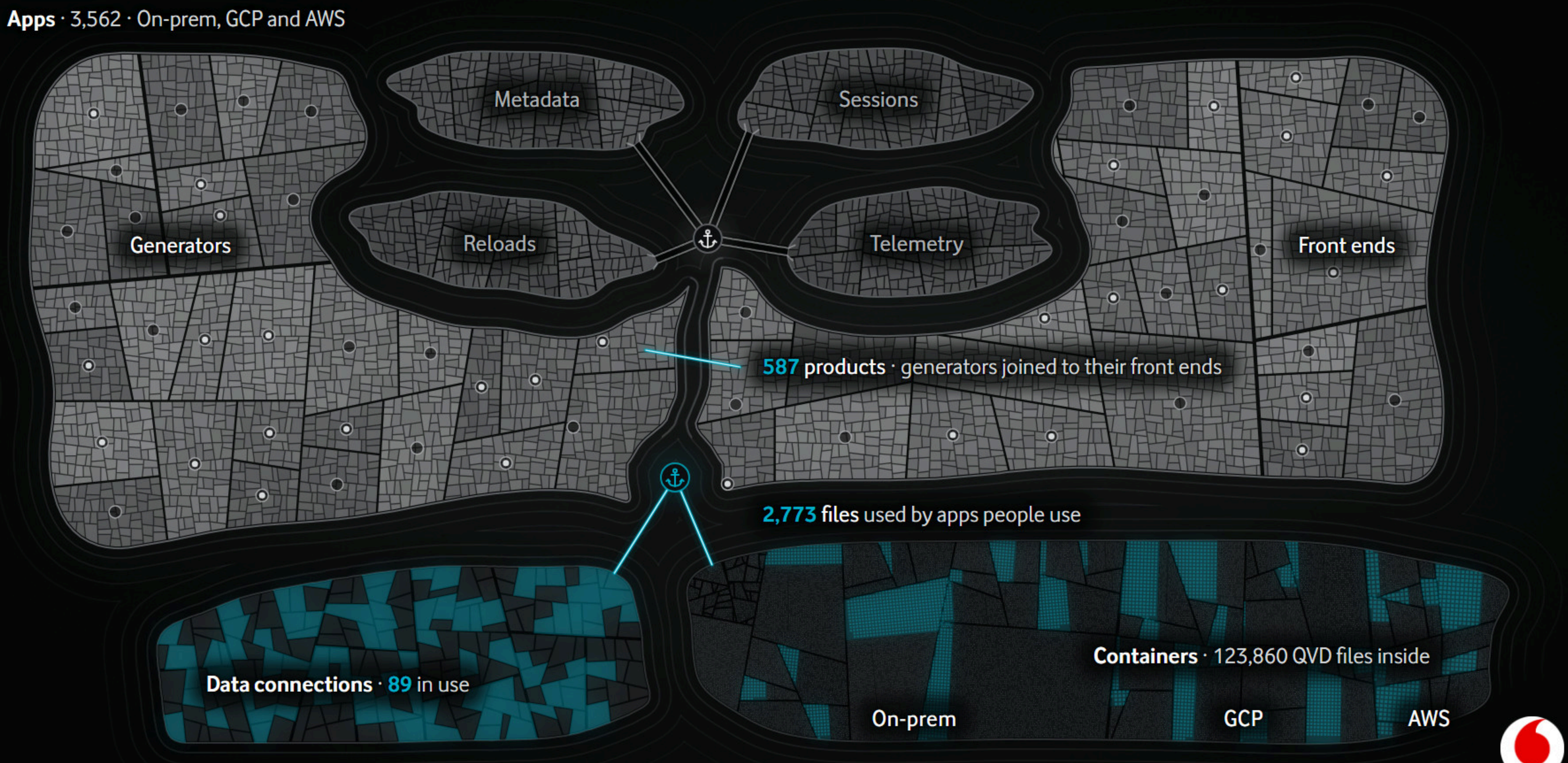
Sales YoY front end · reload log · 12 Oct 2026 · 07:02

```
        Execution started.
0011    LET vPII = 'Yes'
0018    // ---- Data model ----
0020    LOAD * FROM [lib://SalesContainer/QVD/sales_orders.qvd] (qvd)
0024    LOAD * FROM [lib://SalesContainer/QVD/customers.qvd] (qvd)
0028    LOAD * FROM [lib://MasterContainer/QVD/calendar.qvd] (qvd)
```



Illustrative layout · real counts

One clue became a method



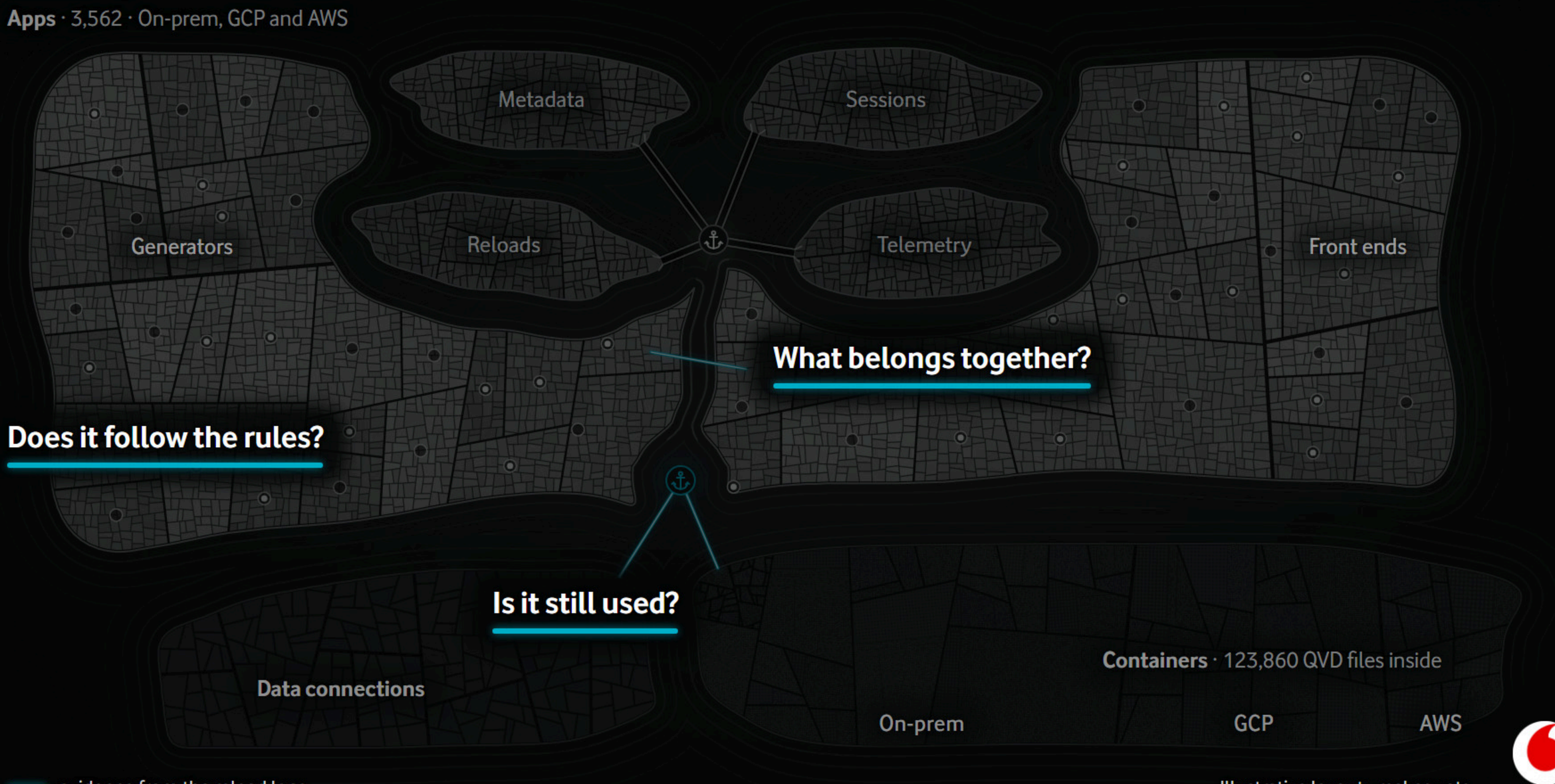
Qlik keeps reload logs for two months · our own copy: four months and growing

Illustrative layout · real counts

Back to our four questions

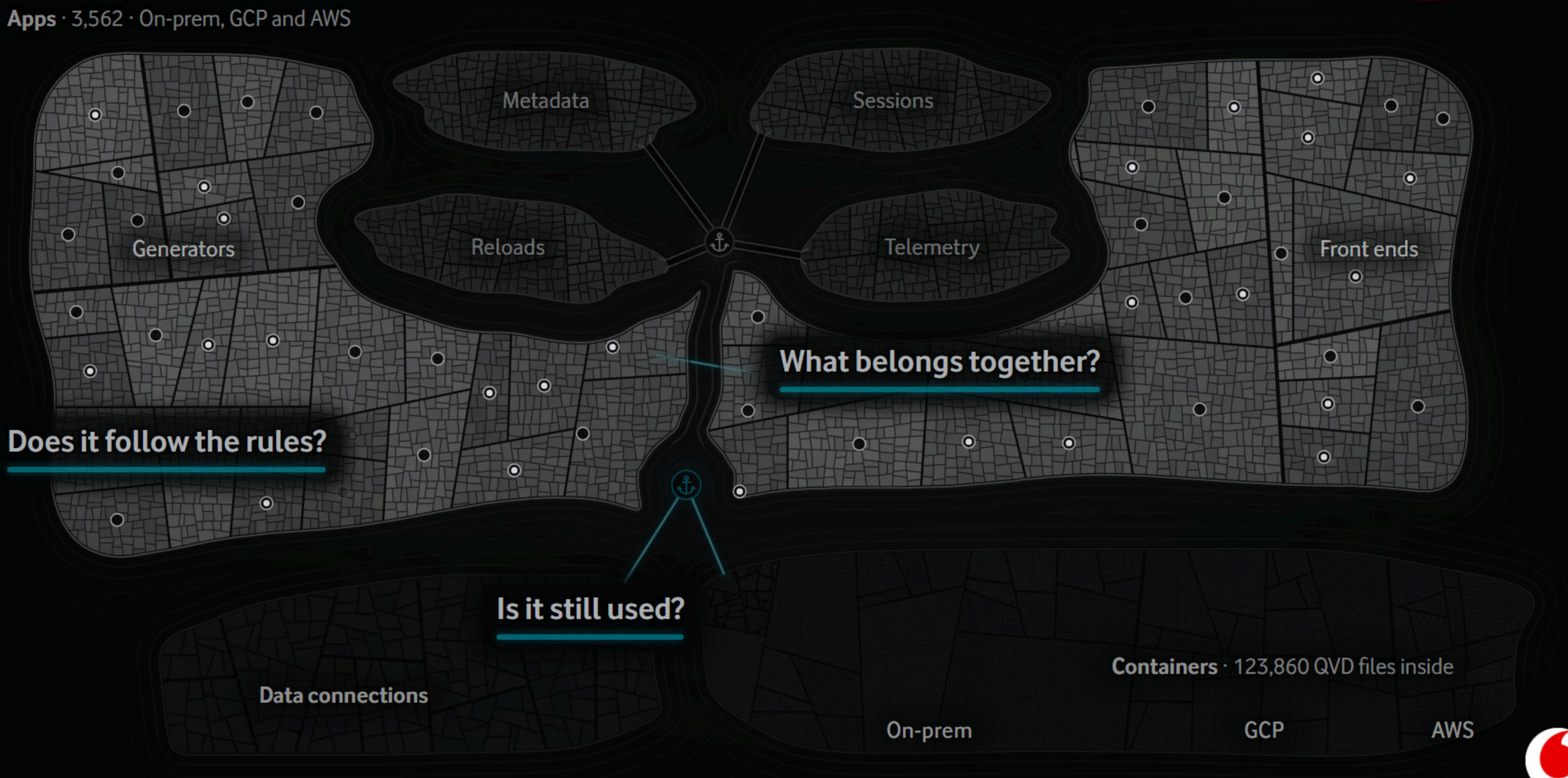
Back to our four questions

Who owns it?



Back to our four questions

last **Who owns it?**



evidence from the reload logs

Illustrative layout · real counts

Is it still used? Data files

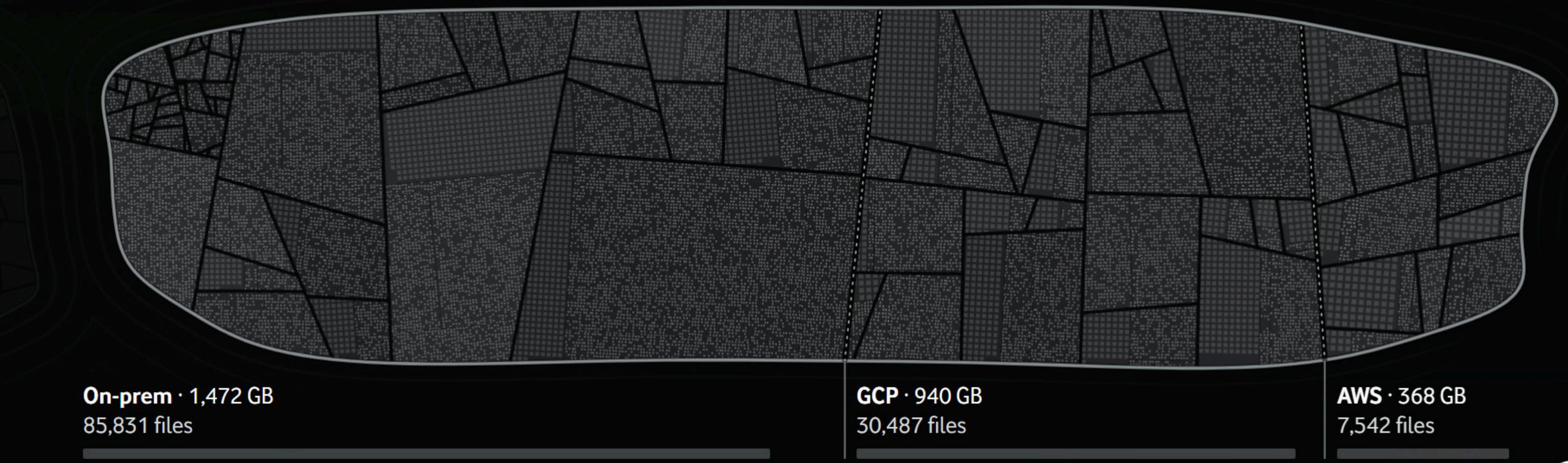
Apps · 3,562 · On-prem, GCP and AWS

Illustrative layout · real counts

Is it still used? Data files

2,780 GB

123,860 QVD files in Production · On-prem, GCP and AWS



On-prem · 1,472 GB
85,831 files

GCP · 940 GB
30,487 files

AWS · 368 GB
7,542 files

Illustrative layout · real counts

Is it still used? Data files

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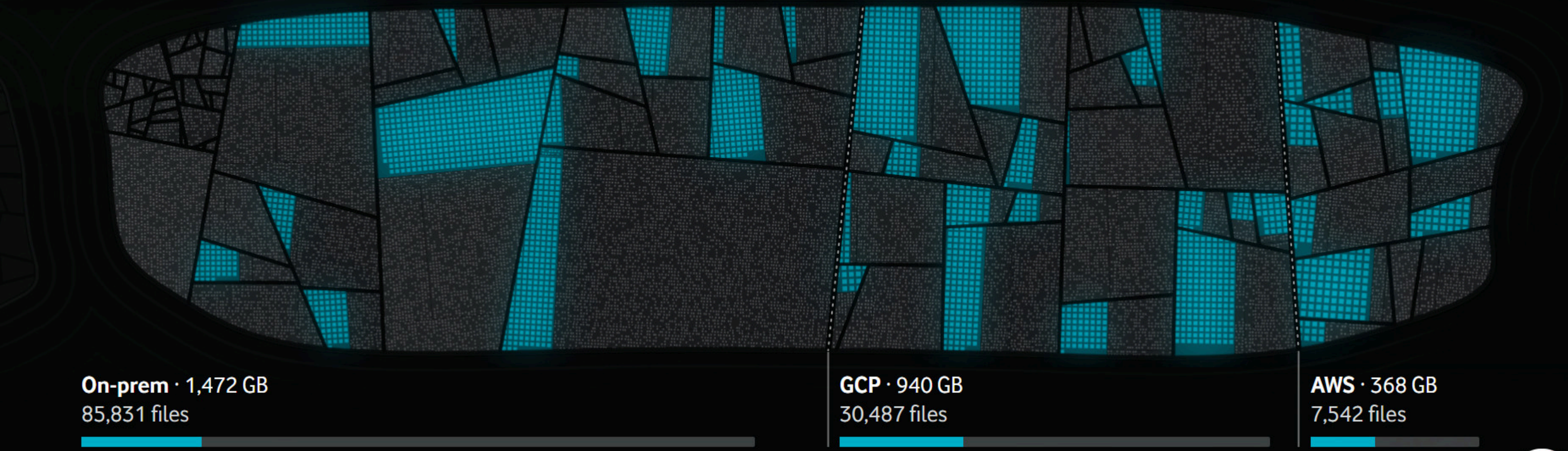
123,860 QVD files in Production · On-prem, GCP and AWS

Used by apps people use

674 GB 24 %

2,773 files 2 %



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Illustrative layout · real counts

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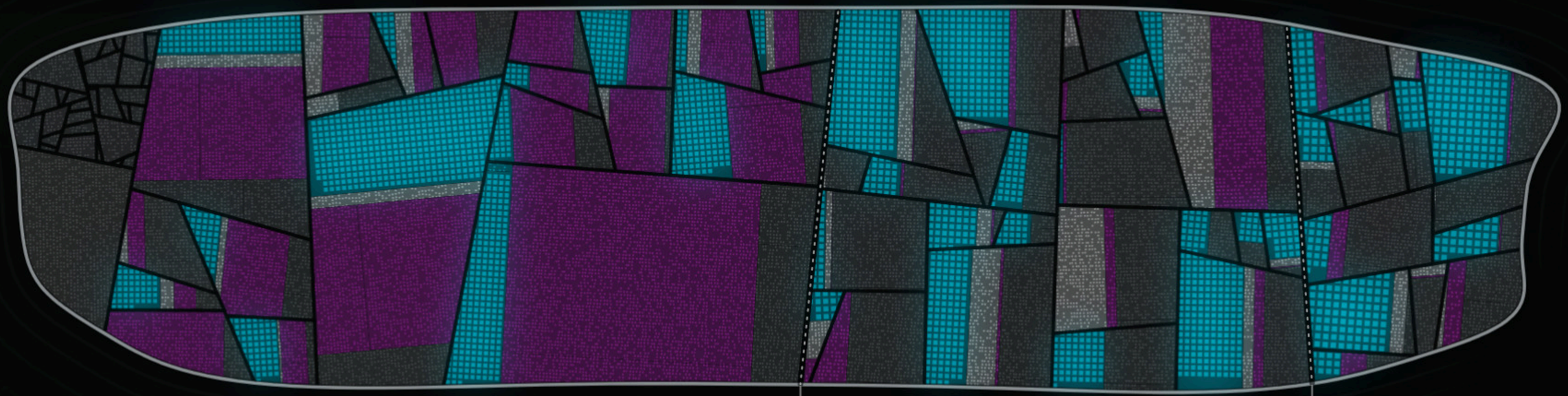
123,860 QVD files in Production · On-prem, GCP and AWS

Used by apps people use	Read only by apps nobody opens	Written, never read	No sign of use
674 GB 24 %	**885 GB** 32 %	**210 GB** 8 %	**1,011 GB** 36 %
2,773 files 2 %	31,247 files 25 %	6,688 files 5 %	83,152 files 67 %



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Illustrative layout · real counts

Is it still used? Data files

1,278 GB

First clean-up round · for owners to review

66,773 files · 488 GB + 790 GB unchanged for 6+ months

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Written, never read
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No sign of use
1,011 GB 36 %
83,152 files 67 %
790 GB unchanged for 6+ months

On-prem · 744.6 of 1,472 GB
51,901 of 85,831 files

GCP · 382.7 of 940 GB
10,570 of 30,487 files

AWS · 150.9 of 368 GB
4,302 of 7,542 files

Illustrative layout · real counts

Is it still used? Data files

1,278 GB

First clean-up round · for owners to review

66,773 files · 488 GB + 790 GB unchanged for 6+ months

100 GB, drawn at the land's scale

The first round is **more than ten times** the space we usually have free

On-prem · 744.6 of 1,472 GB
51,901 of 85,831 files

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A **A, for real**

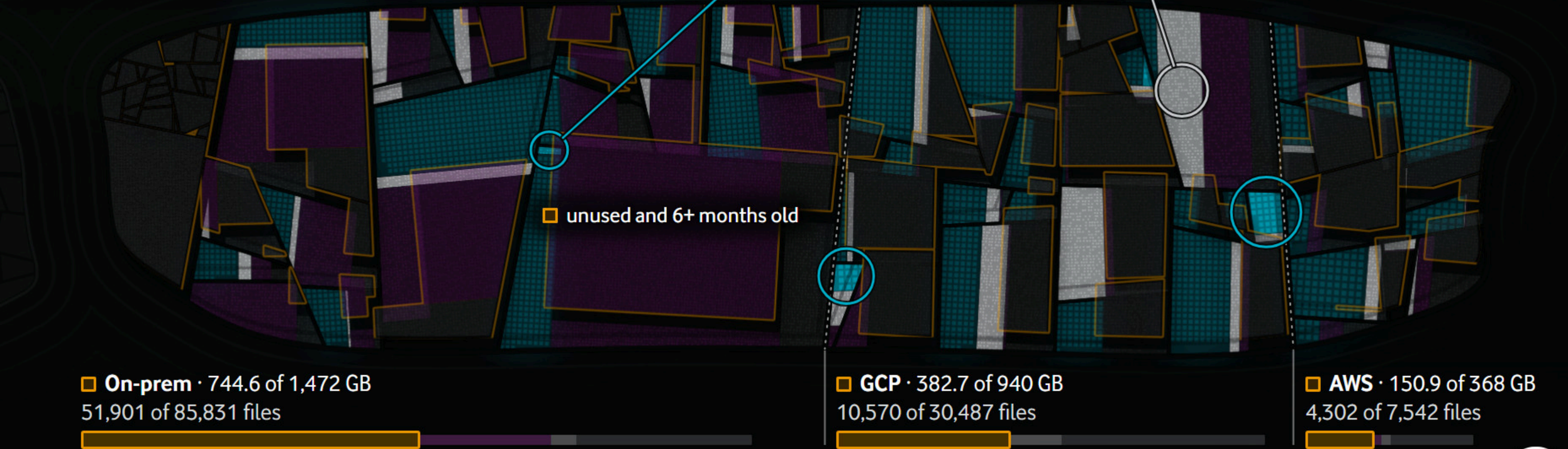
84 files

older than 6 months are still read by live products: deleting by age would break them.

B **B, for real**

210 GB

written and never read, none of it older than 6 months: an age rule would never find them.



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PII

GCP: **471 files** from apps that declare PII data, over 2 years old · for owners to verify

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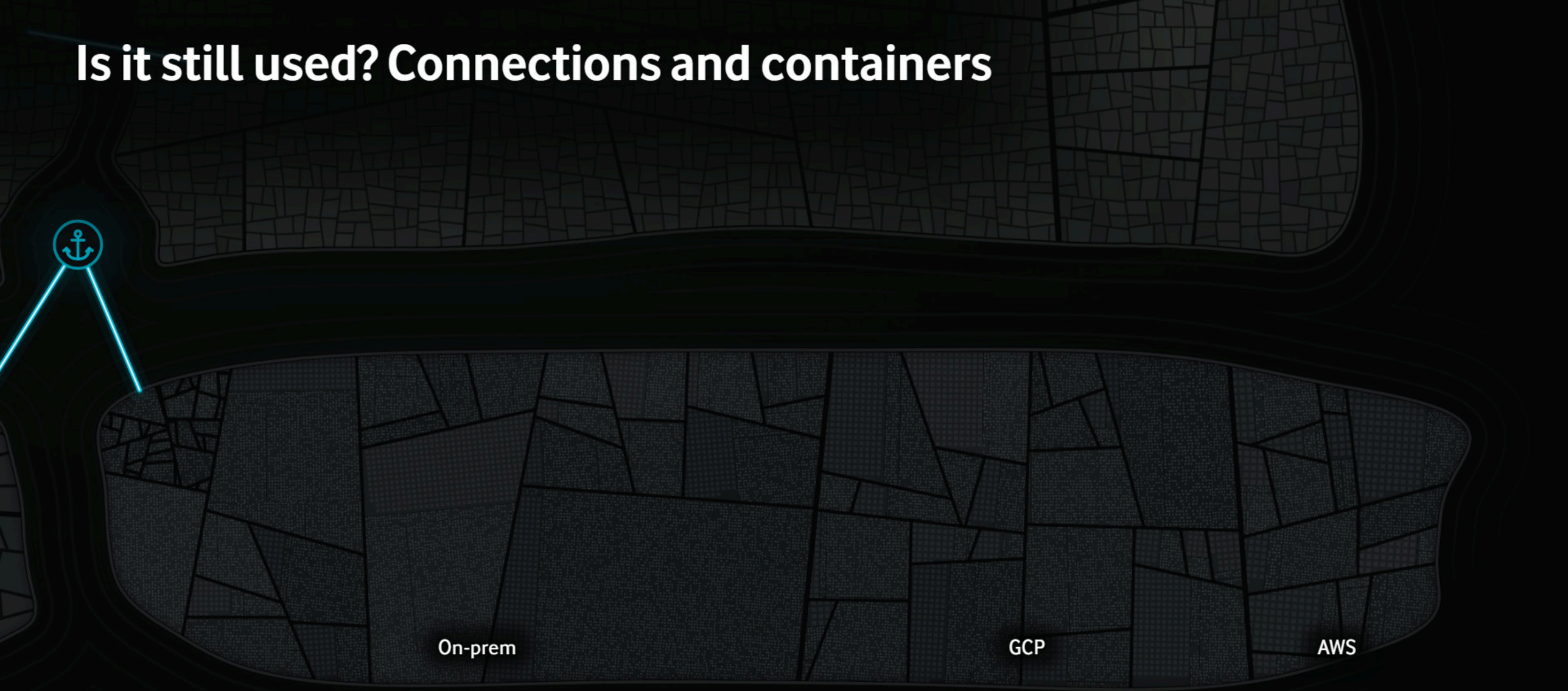
Illustrative layout · real counts

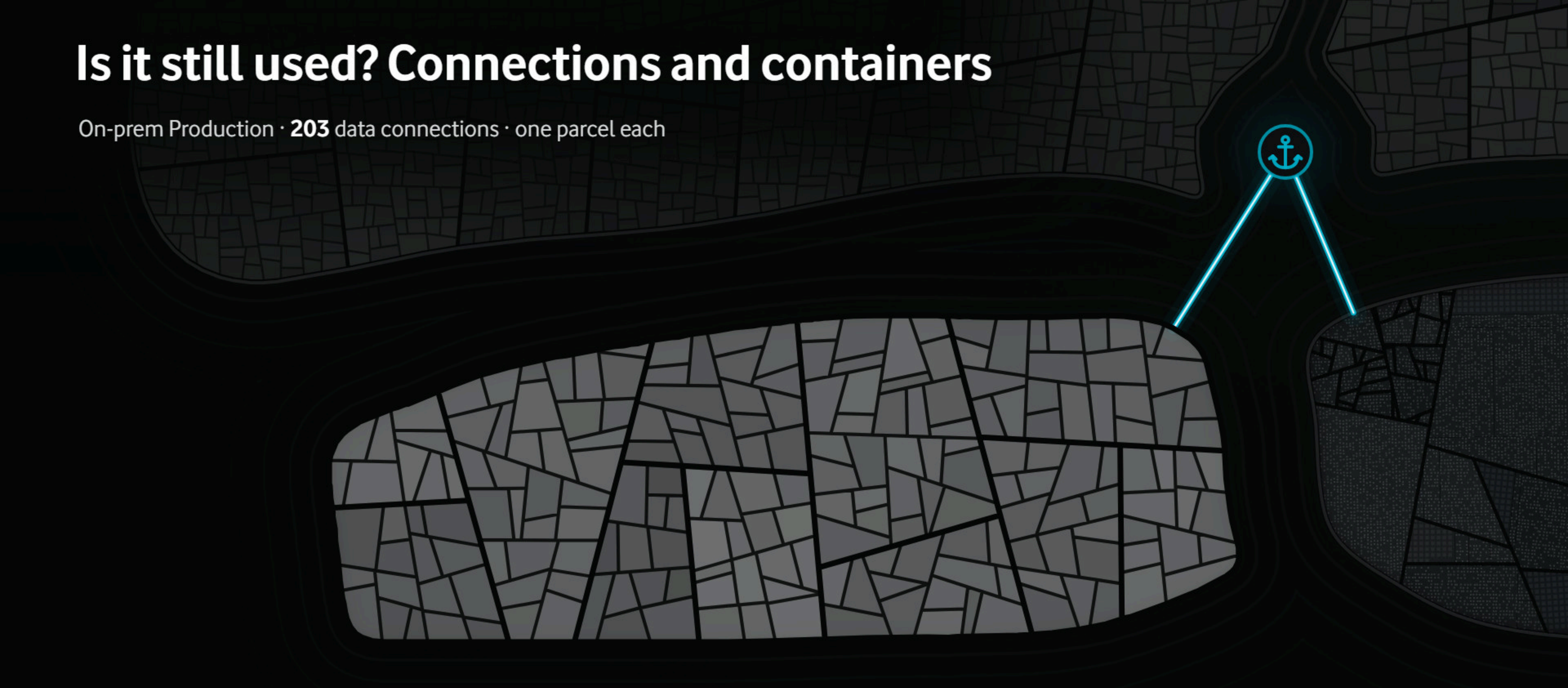
Is it still used? Connections and containers



Illustrative layout · real counts

Is it still used? Connections and containers

On-prem Production · **203** data connections · one parcel each



Illustrative layout · real counts

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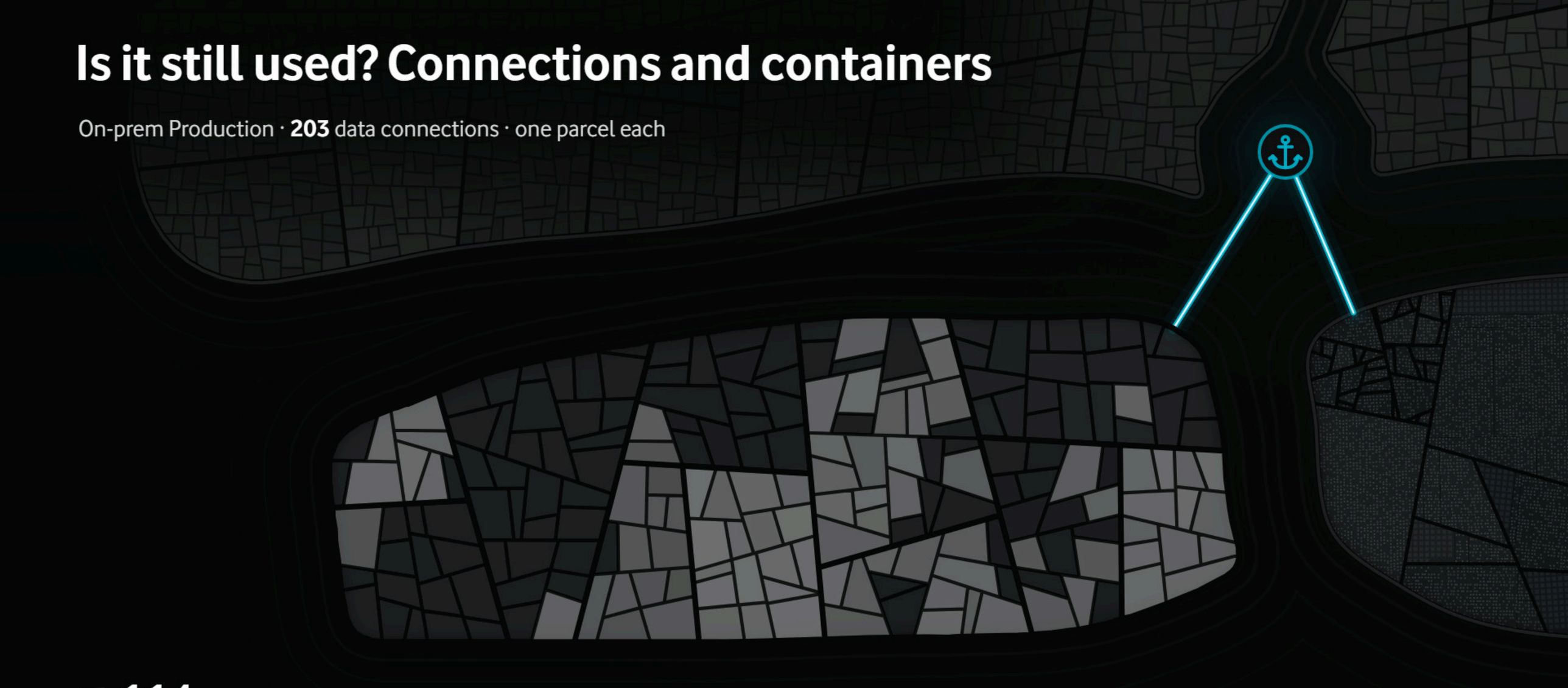

114

unused 3+ months

Illustrative layout · real counts

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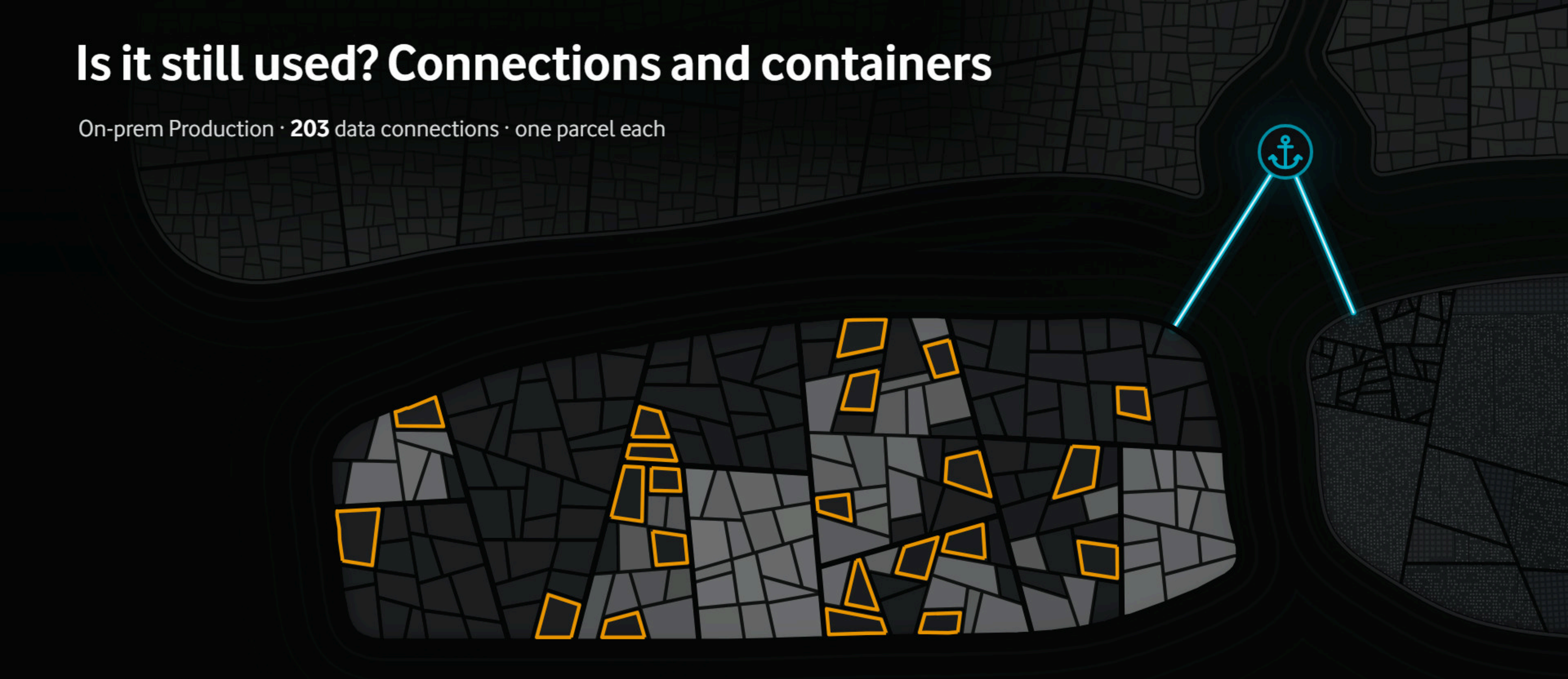

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unused 3+ months

22
feed only apps nobody opens

Illustrative layout · real counts

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Only 67 of 203 are candidates to migrate

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Illustrative layout · real counts

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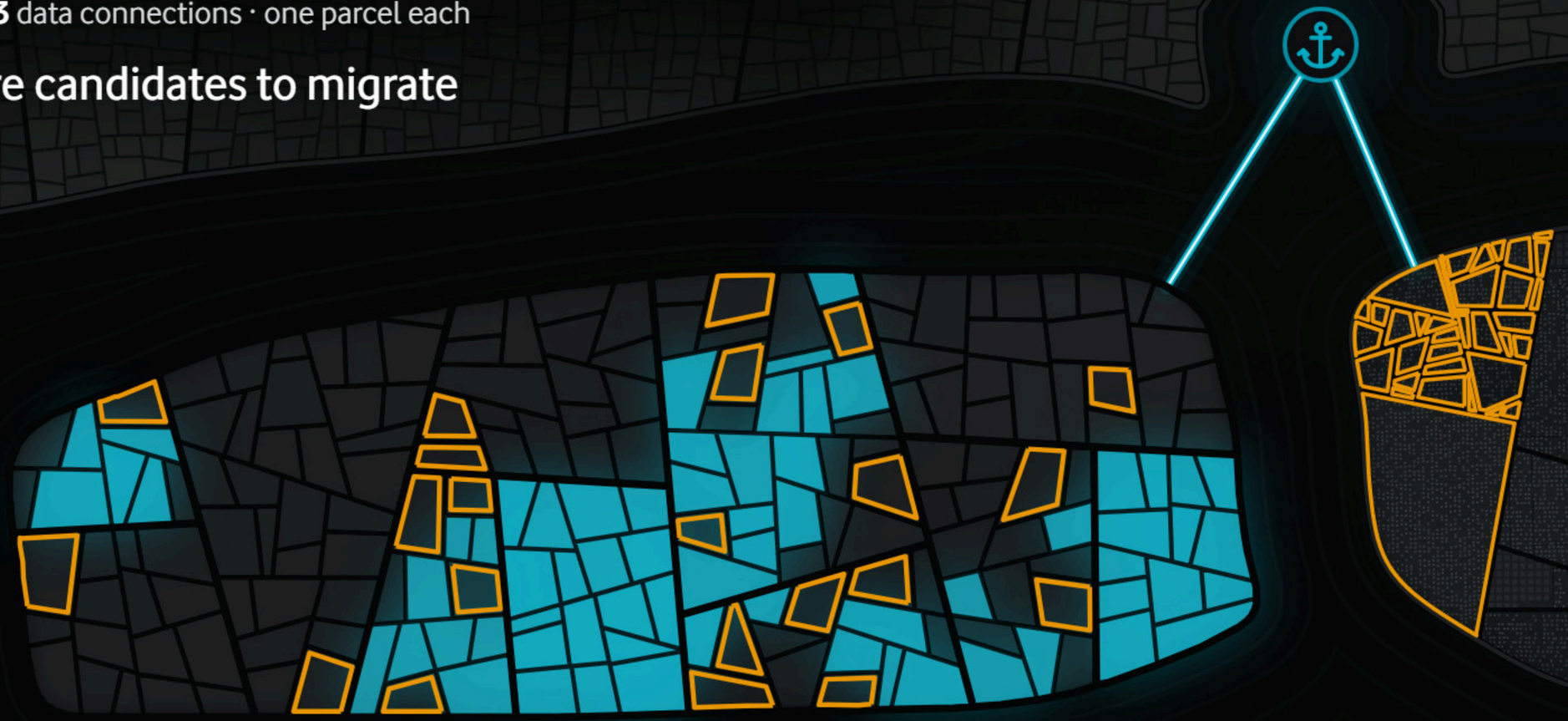


114
unused 3+ months

22
feed only apps nobody opens

67
serve apps in use

37
On-prem containers with no activity

Illustrative layout · real counts

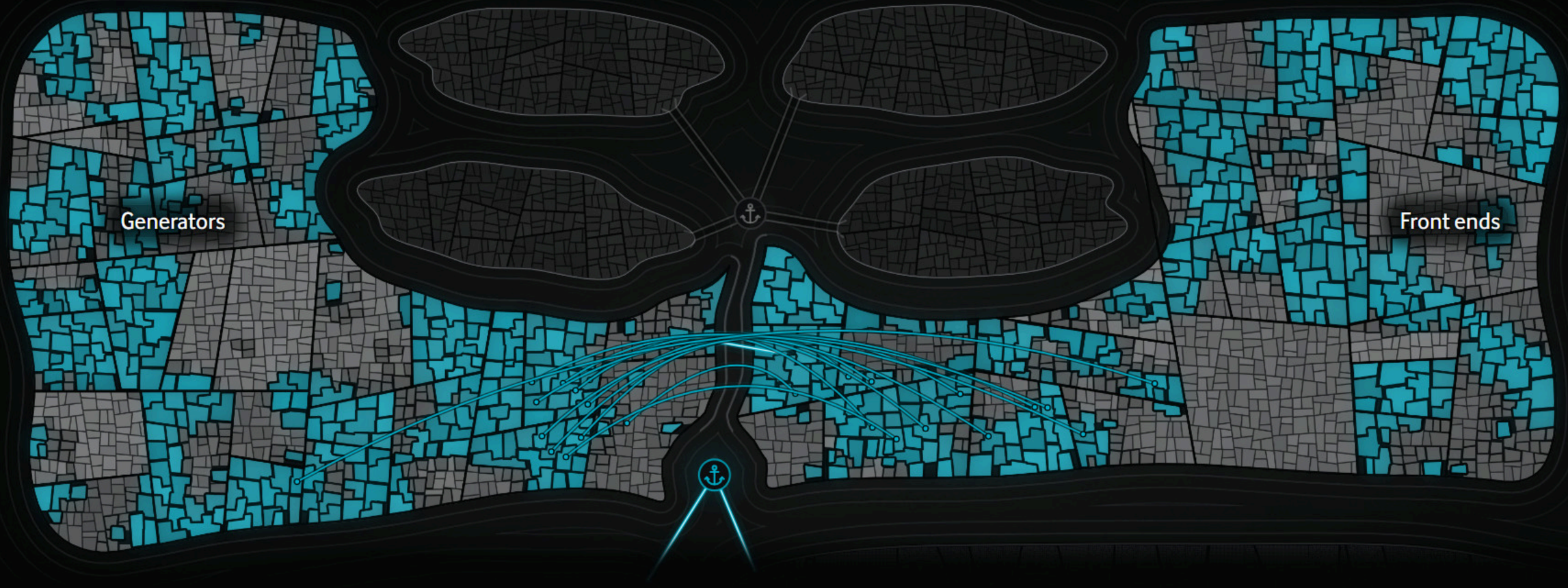
What belongs together? Products

Illustrative layout · real counts

What belongs together? Products

3,562 apps in Production · one parcel each

1,208 products **587** multi-app · 1,779 apps **621** standalone



Illustrative layout · real counts

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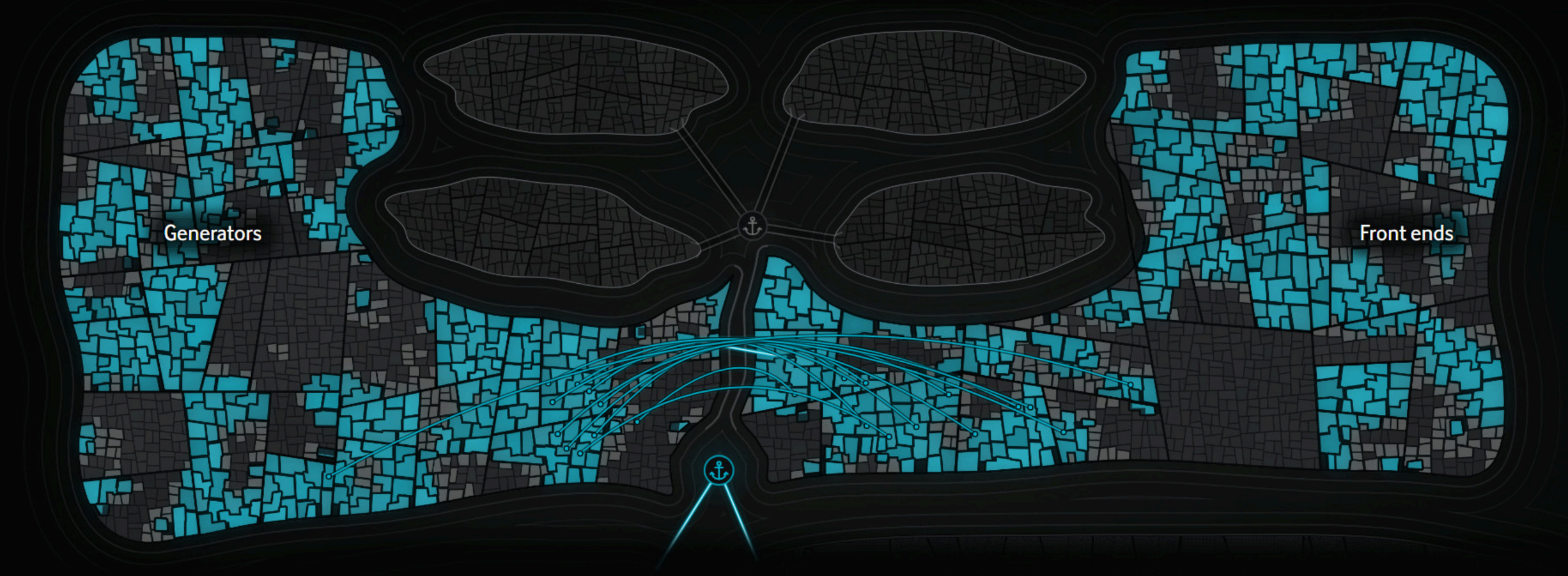

Illustrative layout · real counts

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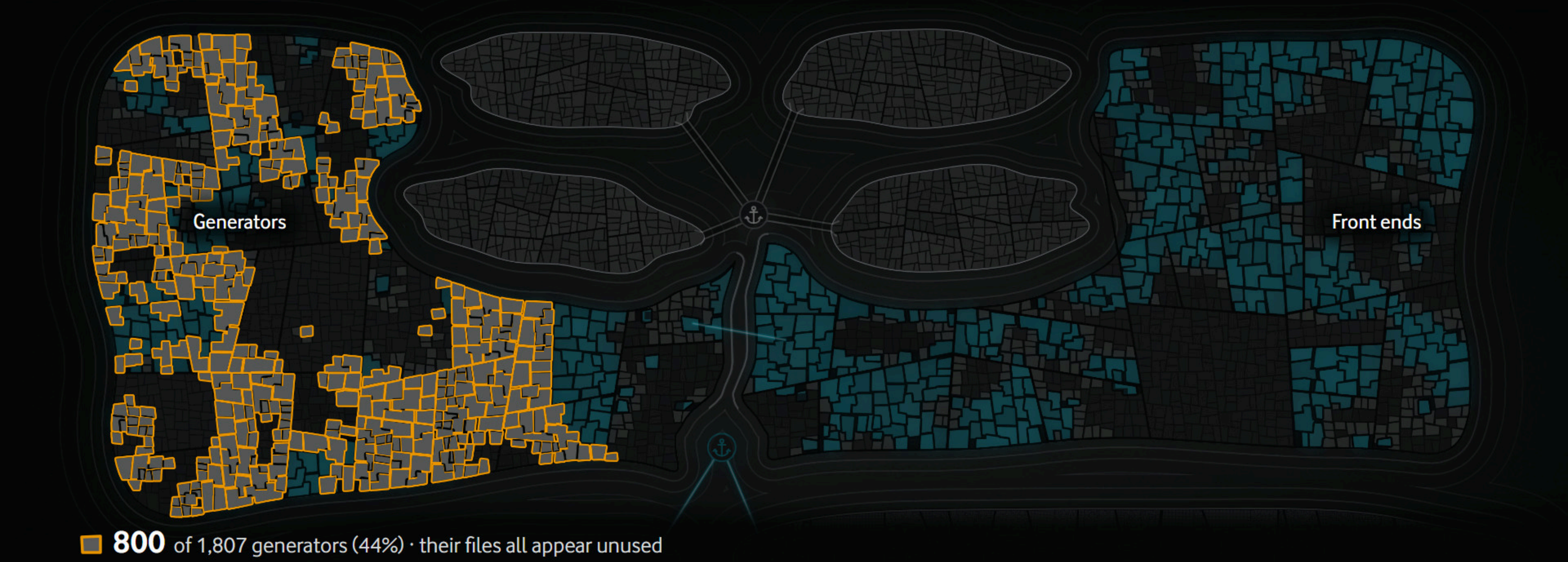


800 of 1,807 generators (44%) · their files all appear unused

Illustrative layout · real counts

What belongs together? Products

3,562 apps in Production · one parcel each

1,162 with no reload in 4 months

1,208 products

587 multi-app · 1,779 apps

621 standalone

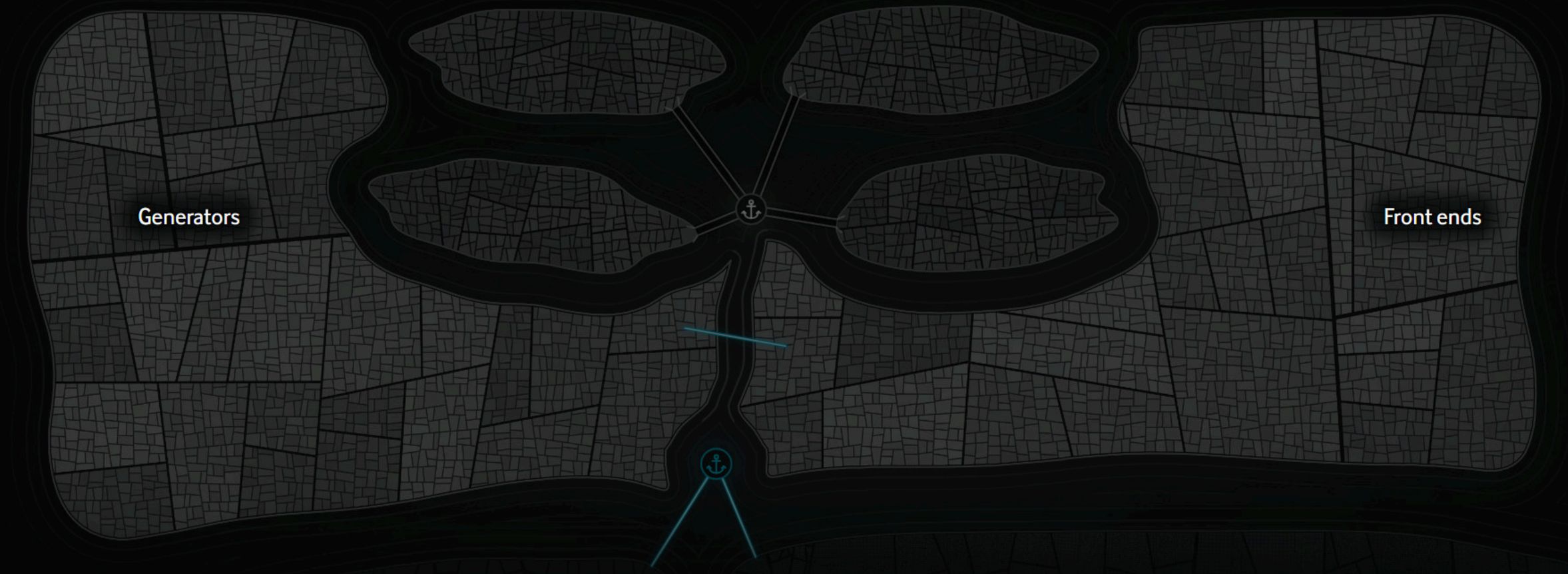
800 of 1,807 generators (44%) · their files all appear unused

September 2026: **126,997** generator reloads · **37,290** (29%) from those generators · about 1,243 a day

Illustrative layout · real counts

Measured as products

Illustrative reload cost, measured **per app**



Illustrative layout · real counts

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Illustrative layout · real counts

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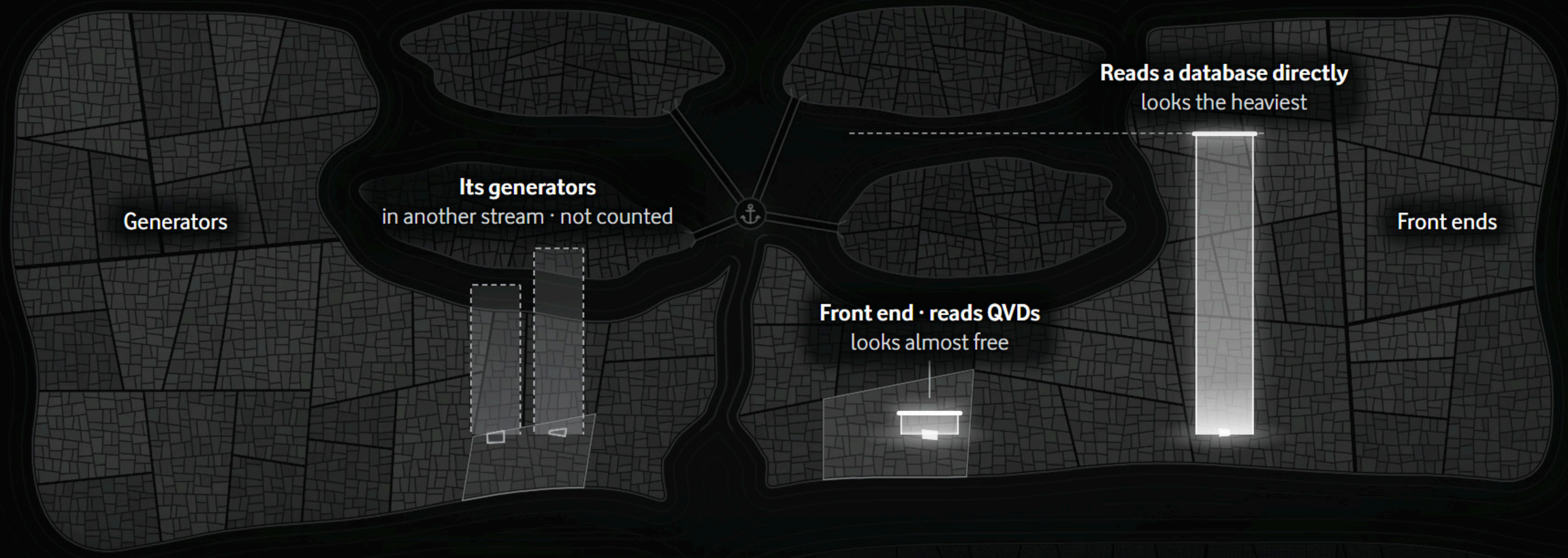

Illustrative layout · real counts

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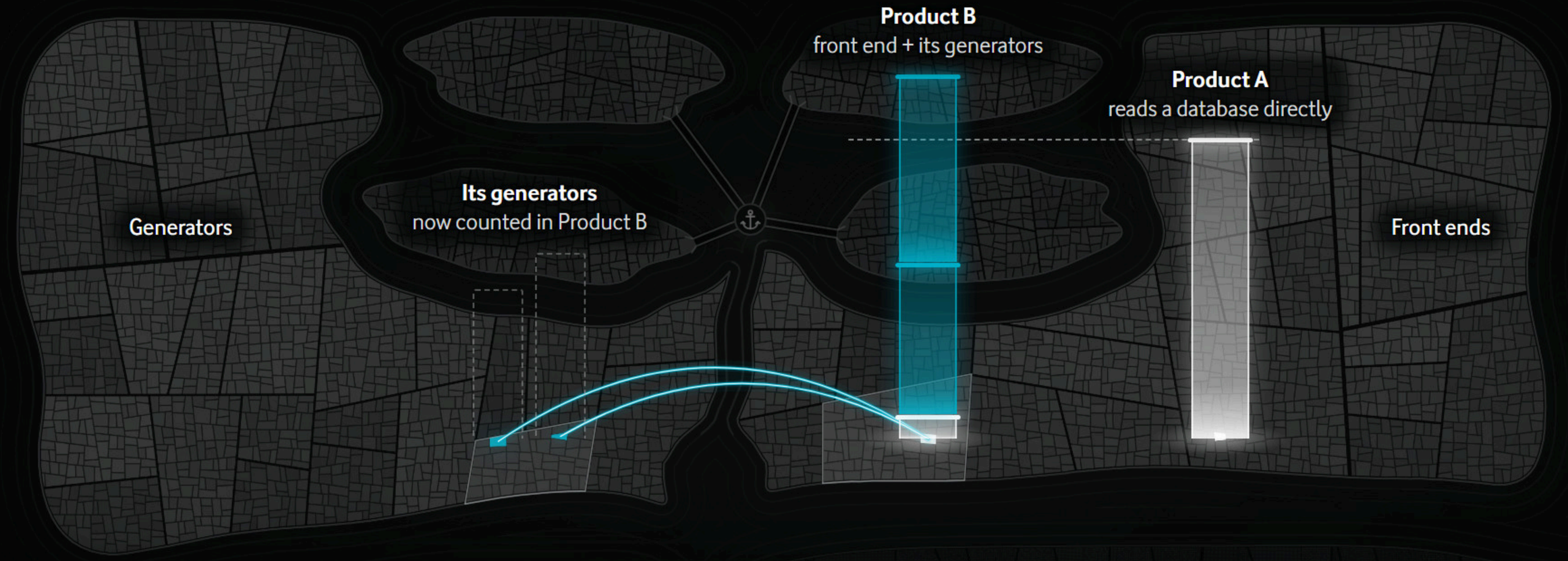
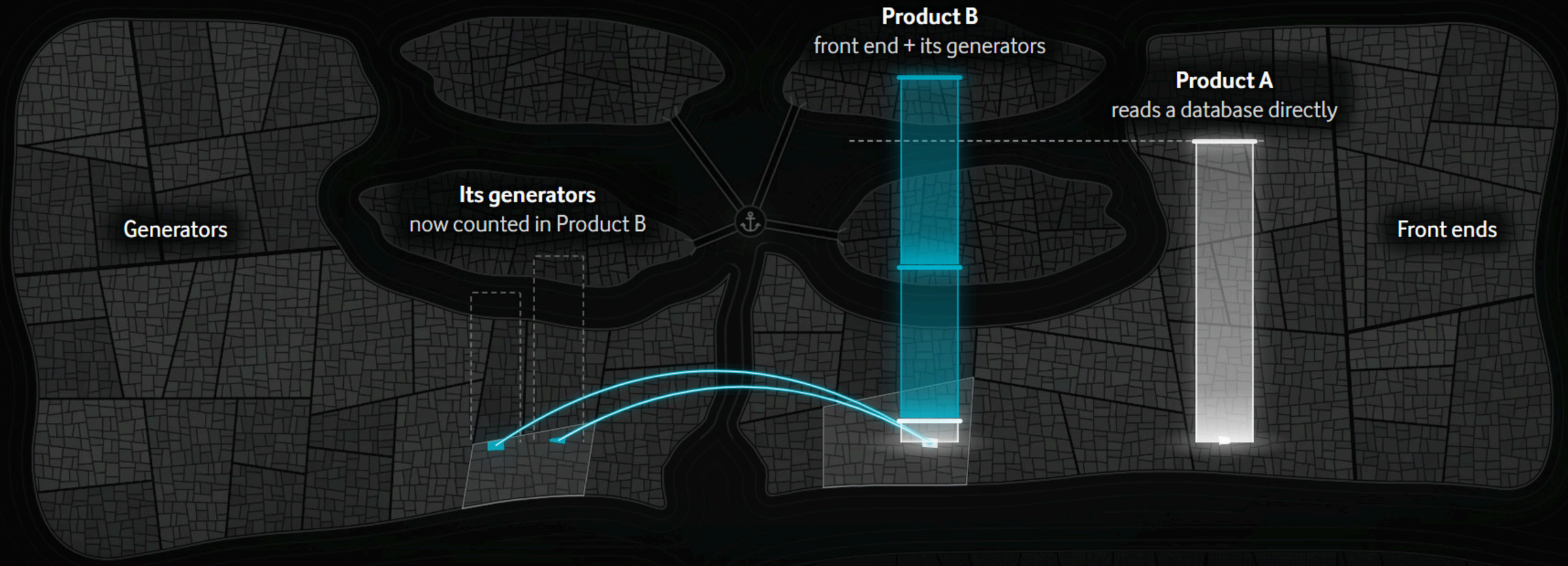


Illustrative layout · real counts

Measured as products

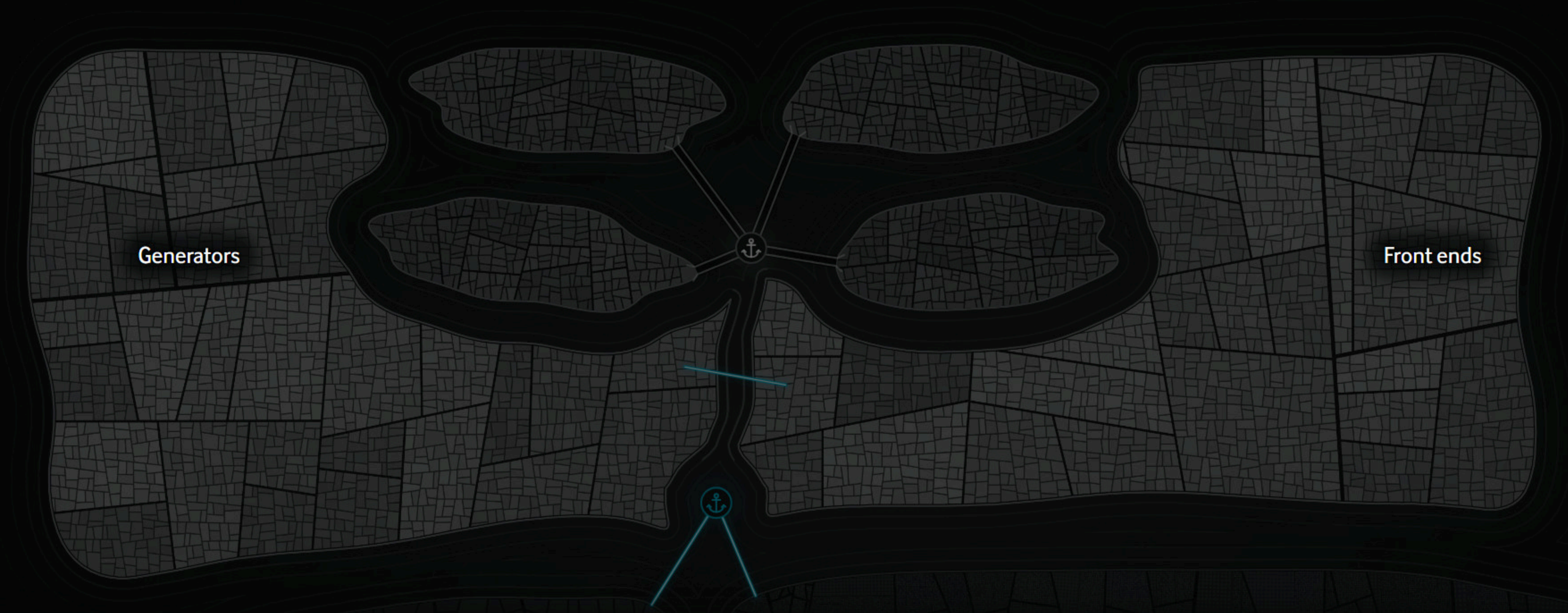
Illustrative reload cost, measured **per product**

We weren't measuring products.
We were measuring pieces of them.



Illustrative layout · real counts

Does it follow the rules?

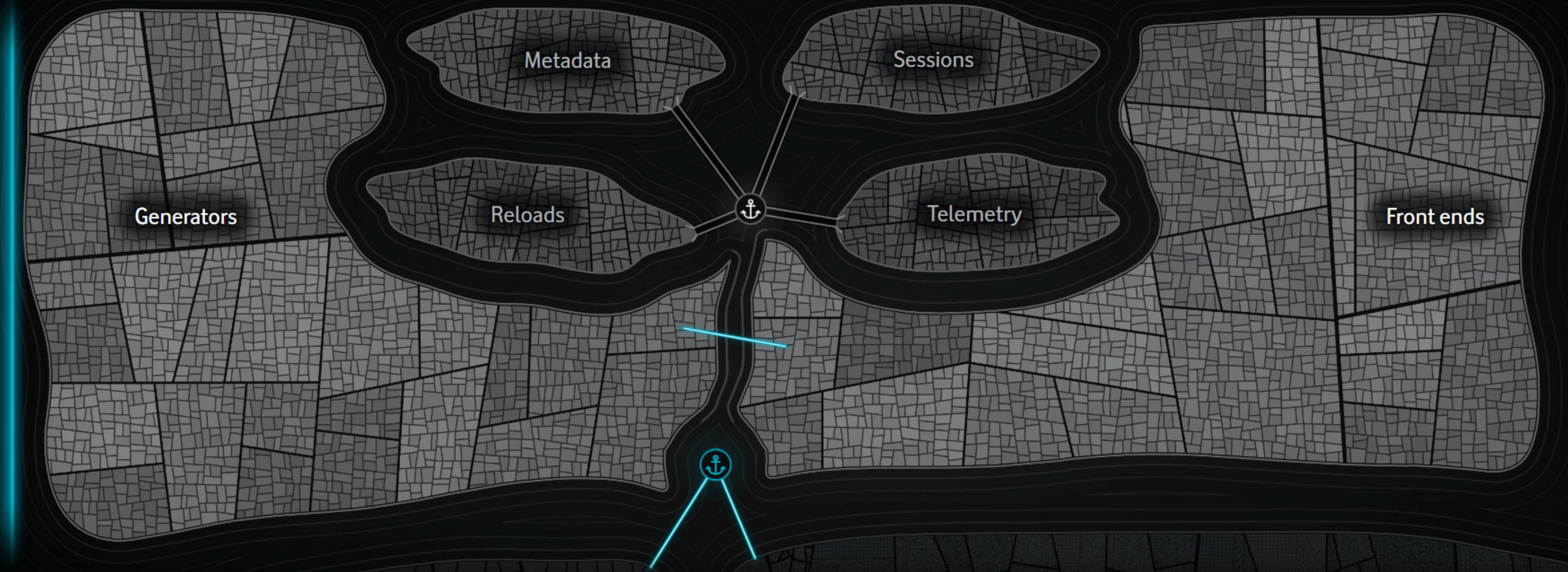


Illustrative layout · real counts

Does it follow the rules?

Template check

PII data · security classification



Illustrative layout · real counts

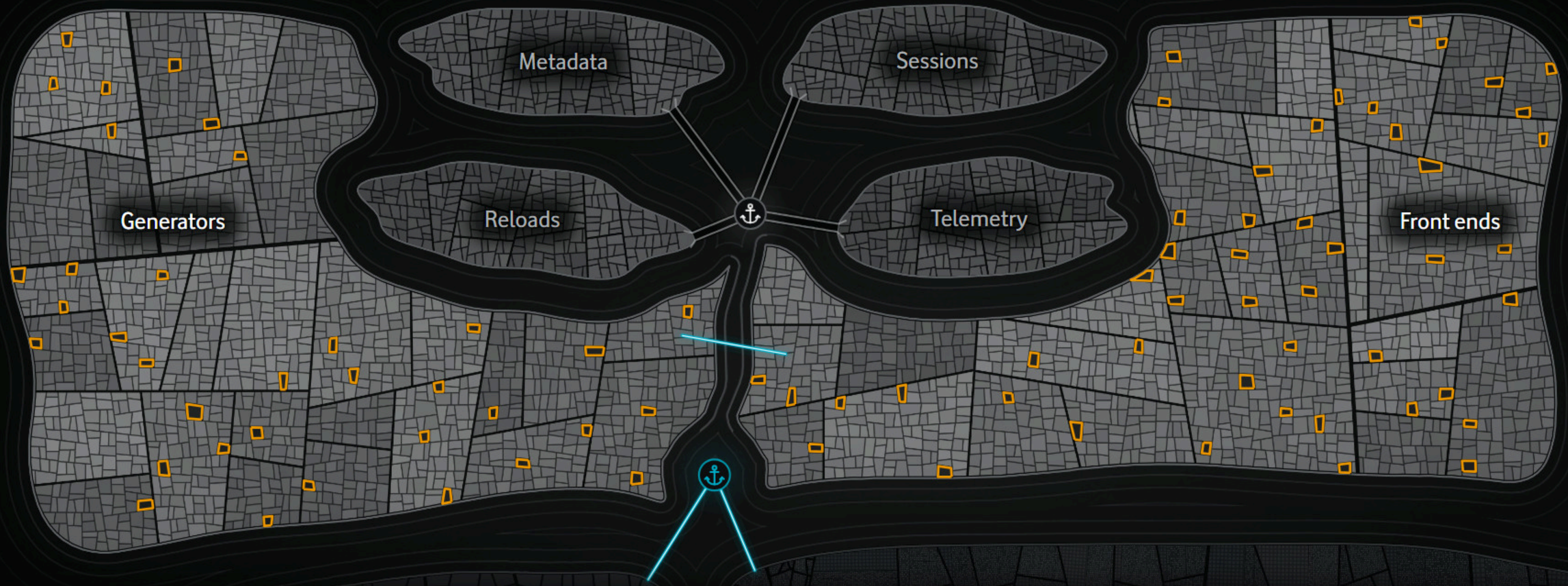
Does it follow the rules?

83 **apps in Production** skip the template

Template check

PII data · security classification

declared as variables in the reload logs · checked app by app



Illustrative layout · real counts

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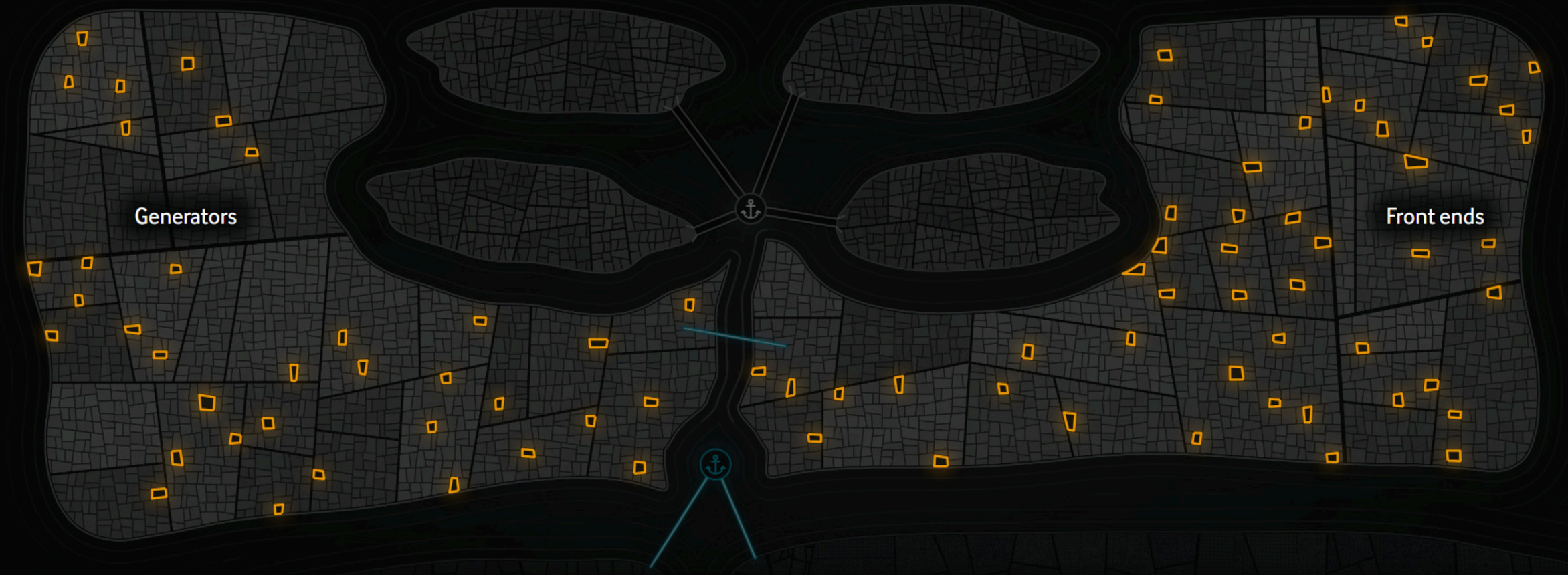
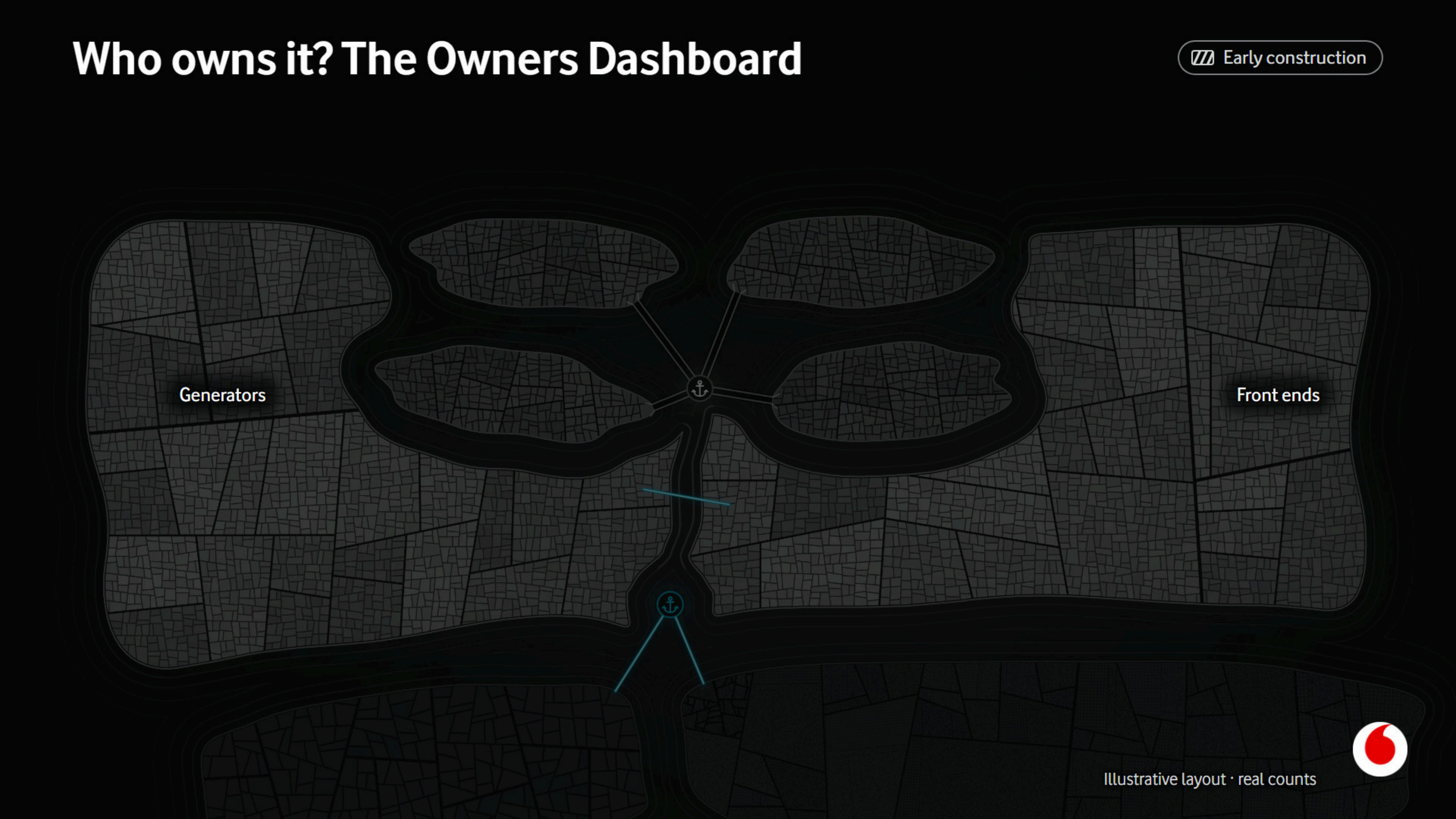


Illustrative layout · real counts

Who owns it? The Owners Dashboard

Early construction



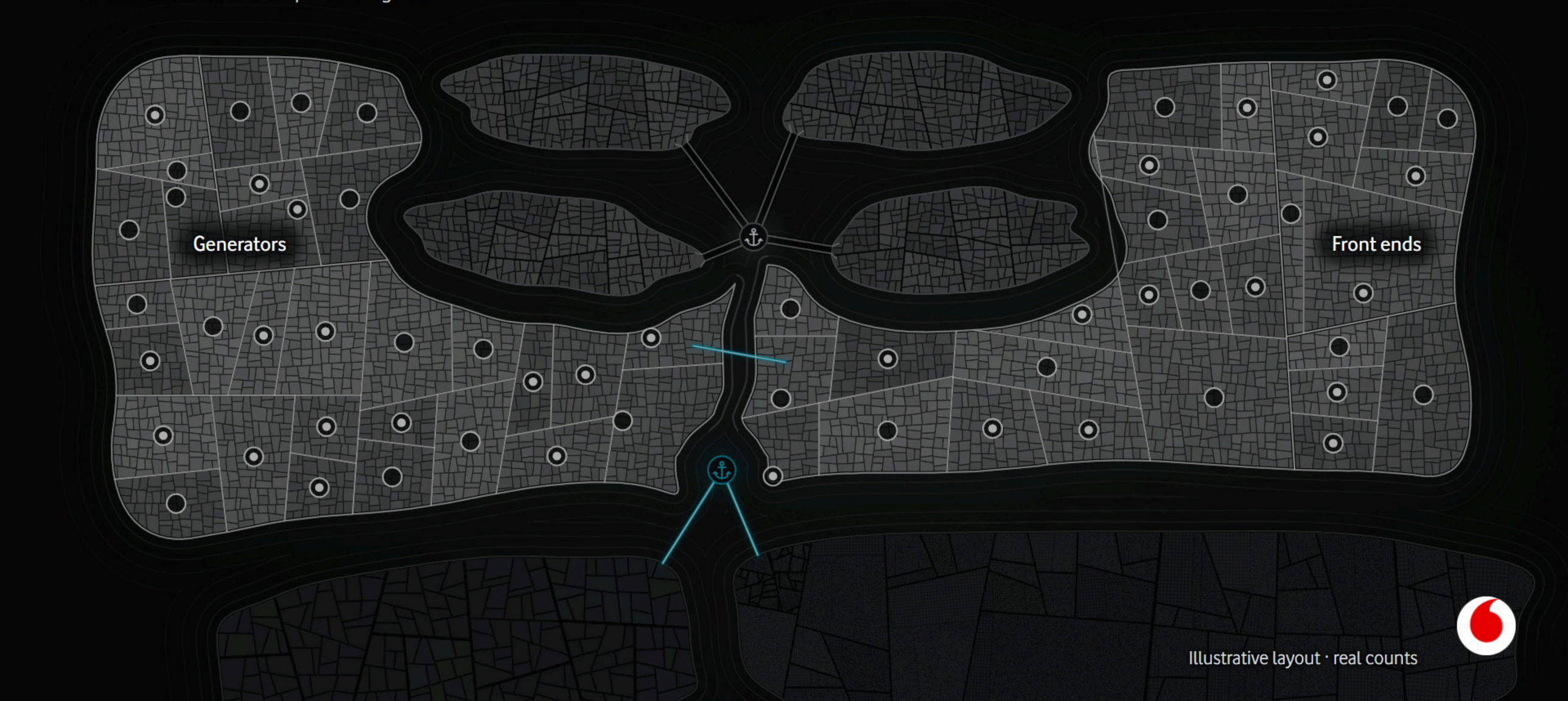
Illustrative layout · real counts

Who owns it? The Owners Dashboard

Early construction

Every stream gets one accountable owner

◉ owner named ○ Ops is finding one



Illustrative layout · real counts

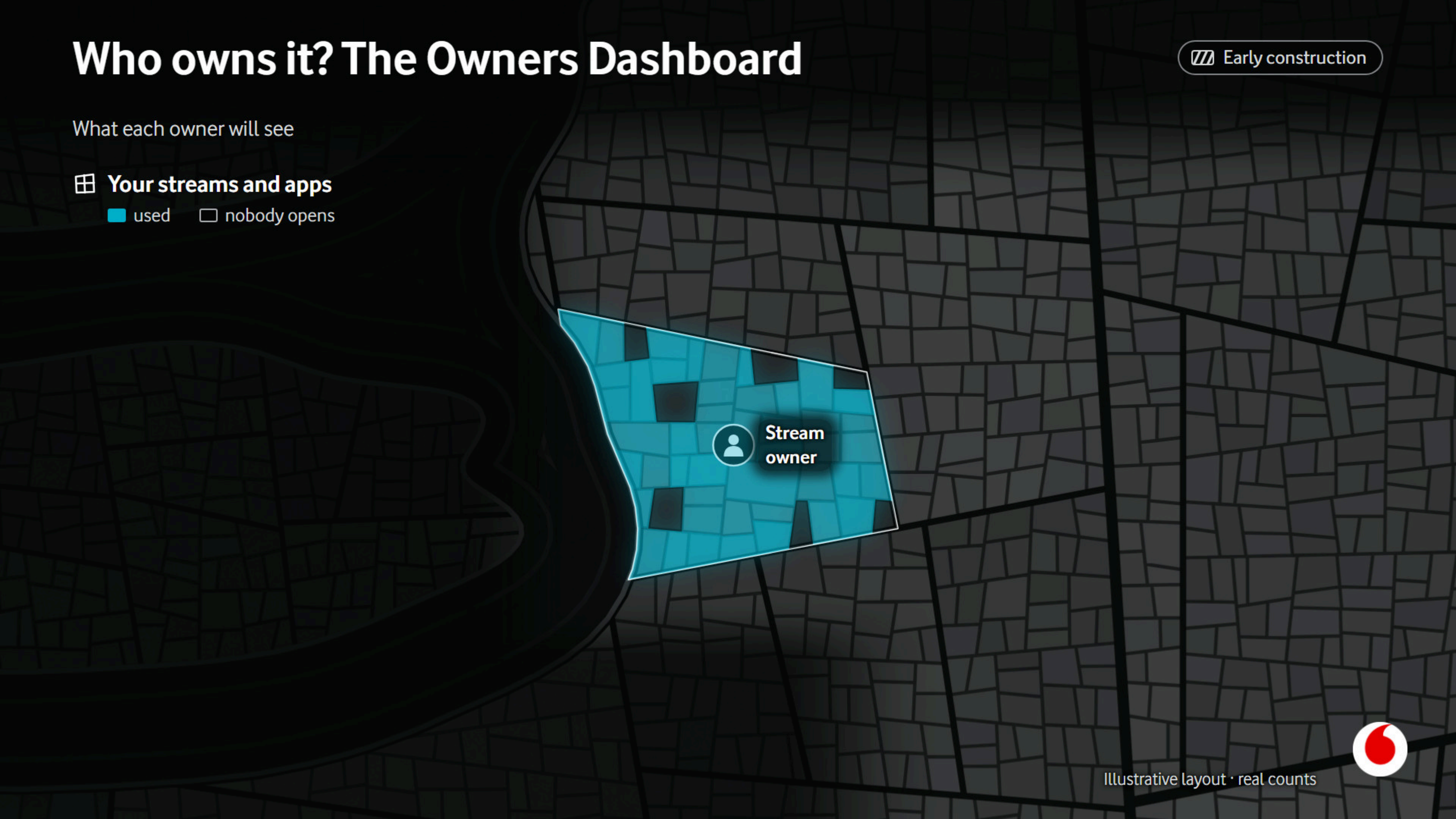
Who owns it? The Owners Dashboard

Early construction

What each owner will see

Your streams and apps

used nobody opens



Illustrative layout · real counts

Who owns it? The Owners Dashboard

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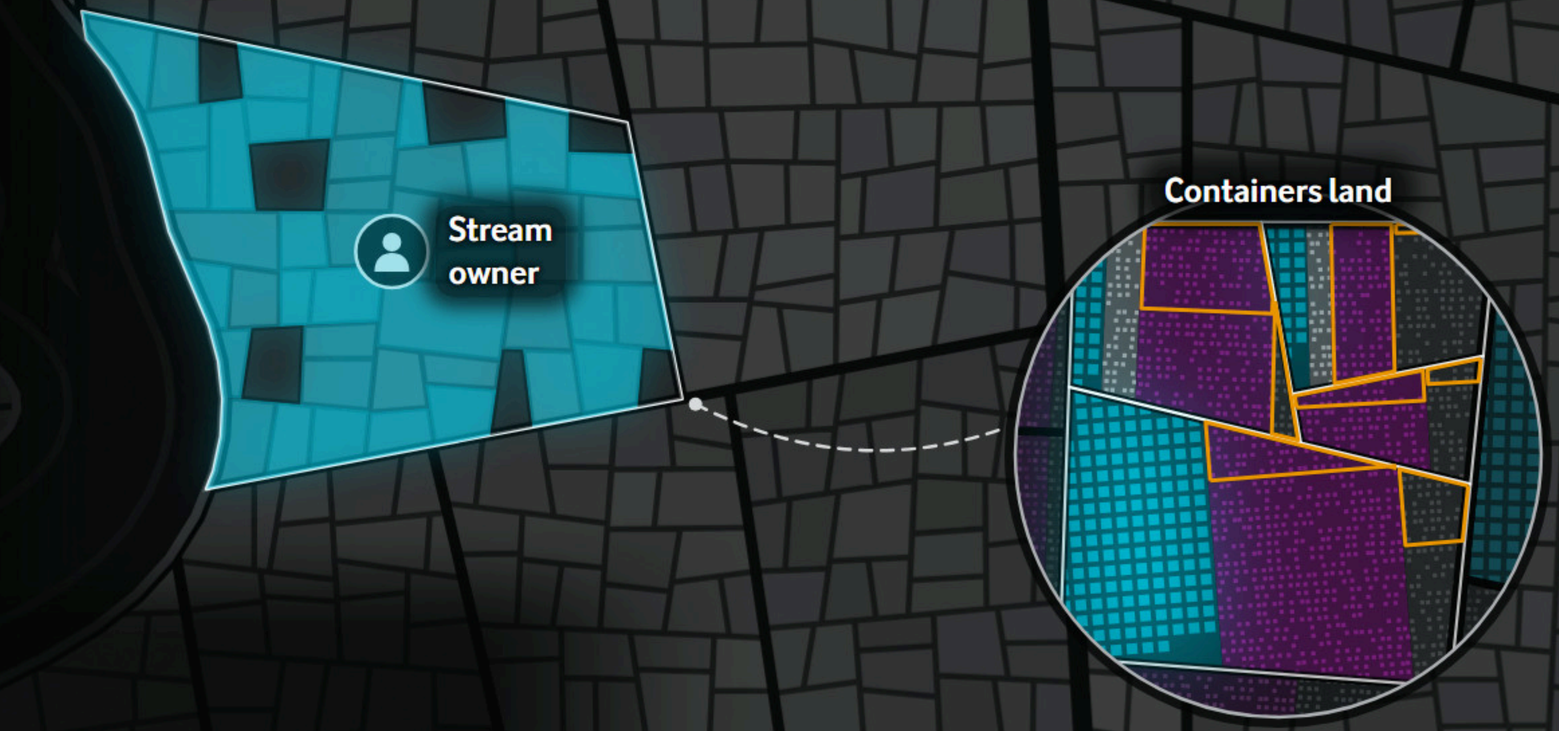
used unused to review



Illustrative layout · real counts

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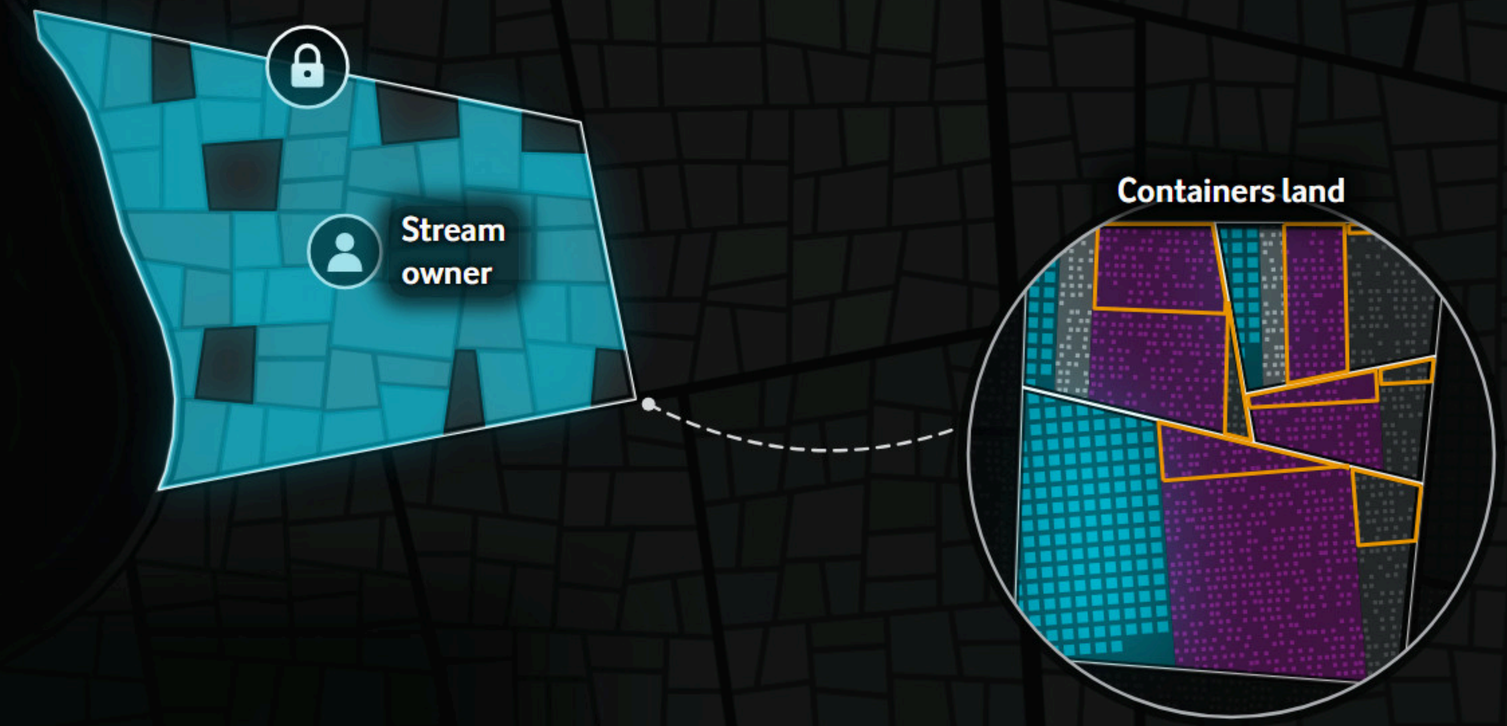
Only your own streams

section access



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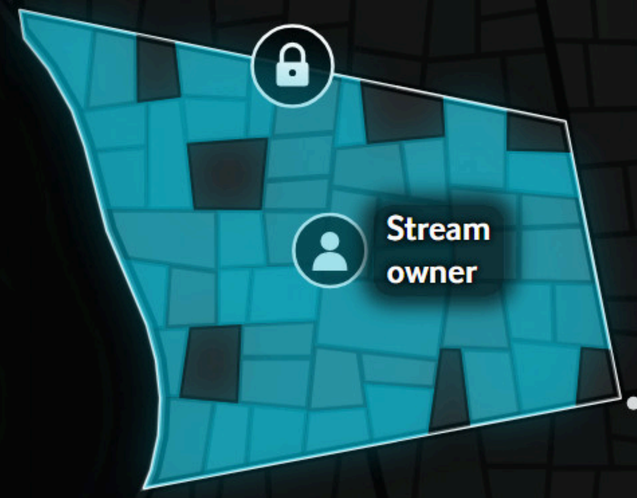

Monthly email · to each owner

Your recommended action items

- **7** apps nobody opens
- Files to review in **4** containers
- **1** generator whose files all appear unused

Illustrative counts

Illustrative layout · real counts

Who owns it? The Owners Dashboard

Early construction

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used nobody opens

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used unused to review

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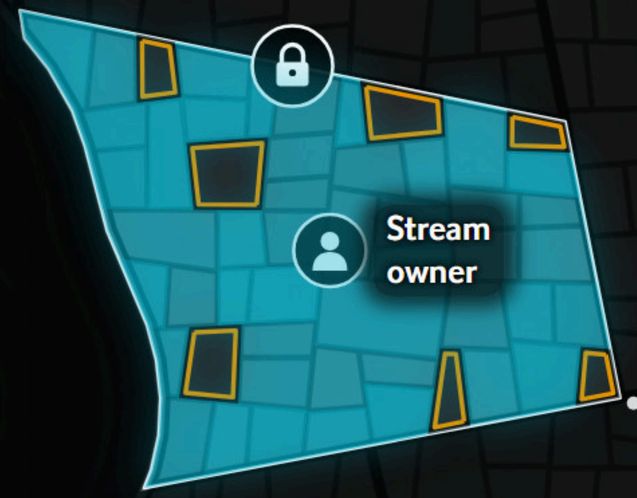

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- **7** apps nobody opens
- Files to review in **4** containers
- **1** generator whose files all appear unused

Illustrative counts

The dashboard won't make decisions for owners.

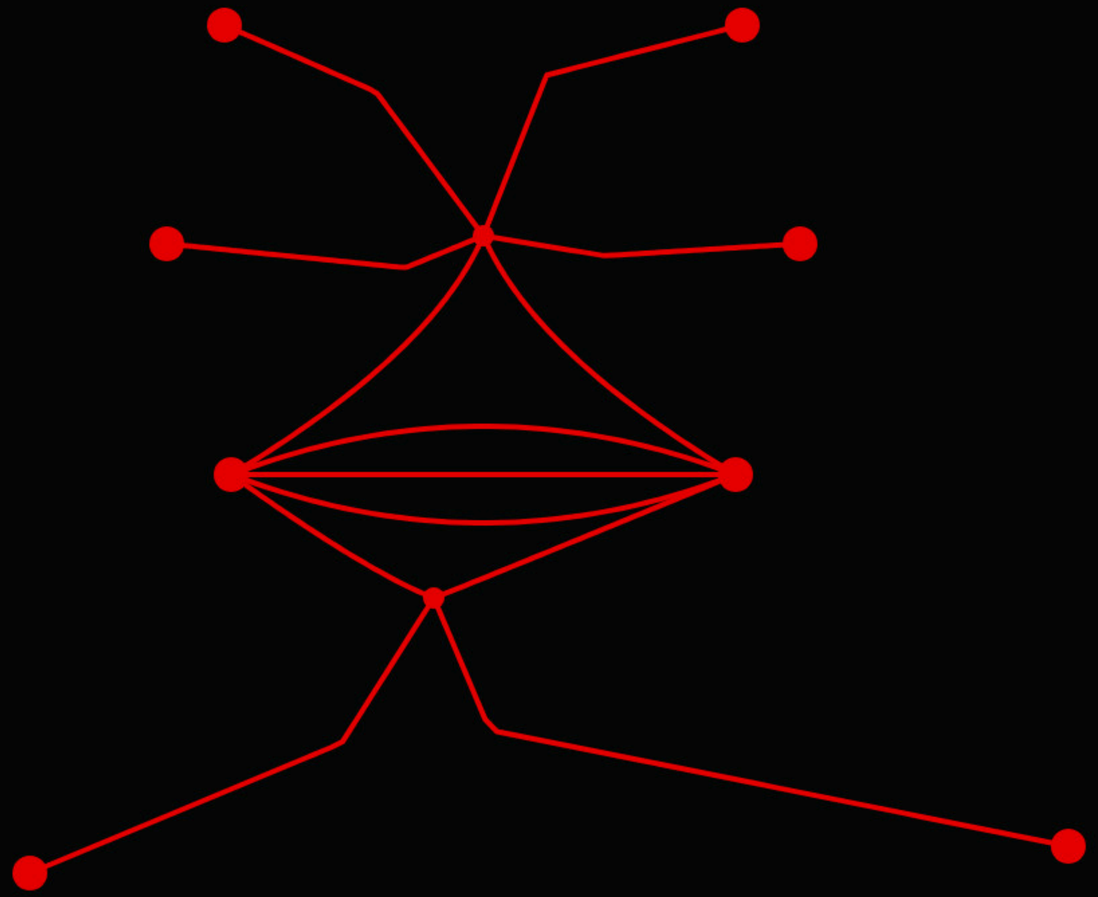
It makes the missing decisions visible.

Illustrative layout · real counts

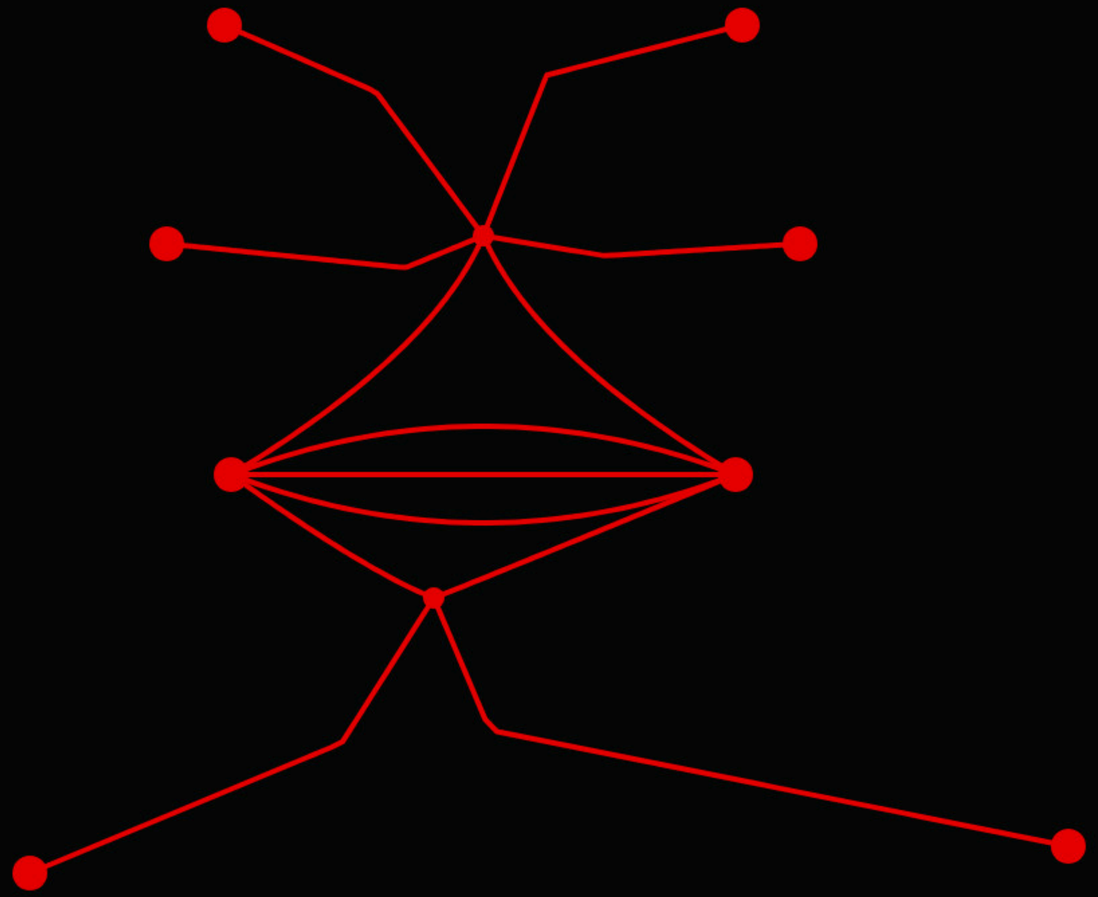
The map we didn't have

Illustrative layout · real counts

The map we didn't have

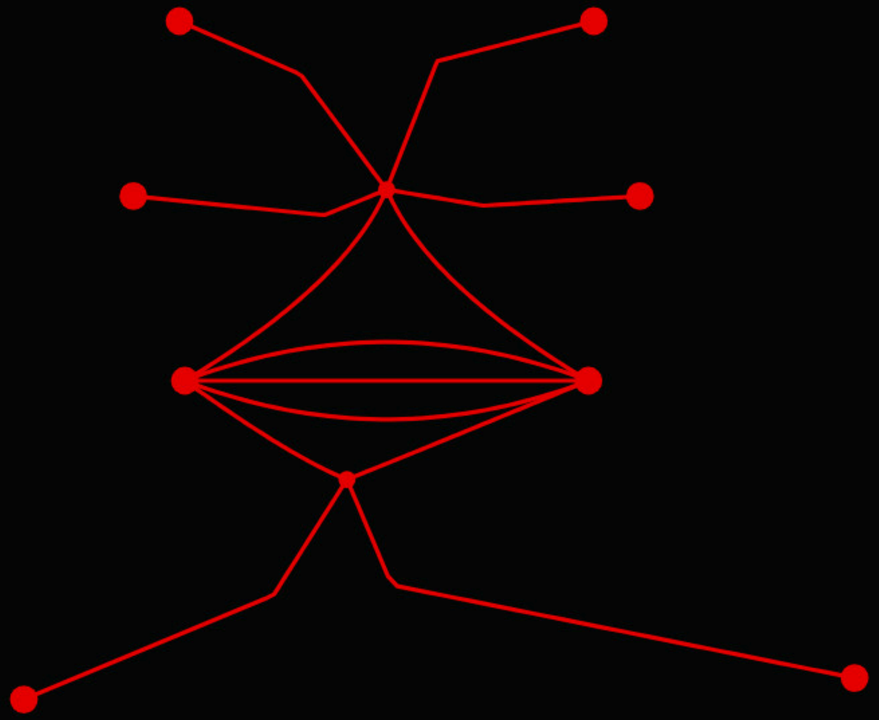


Dots and lines



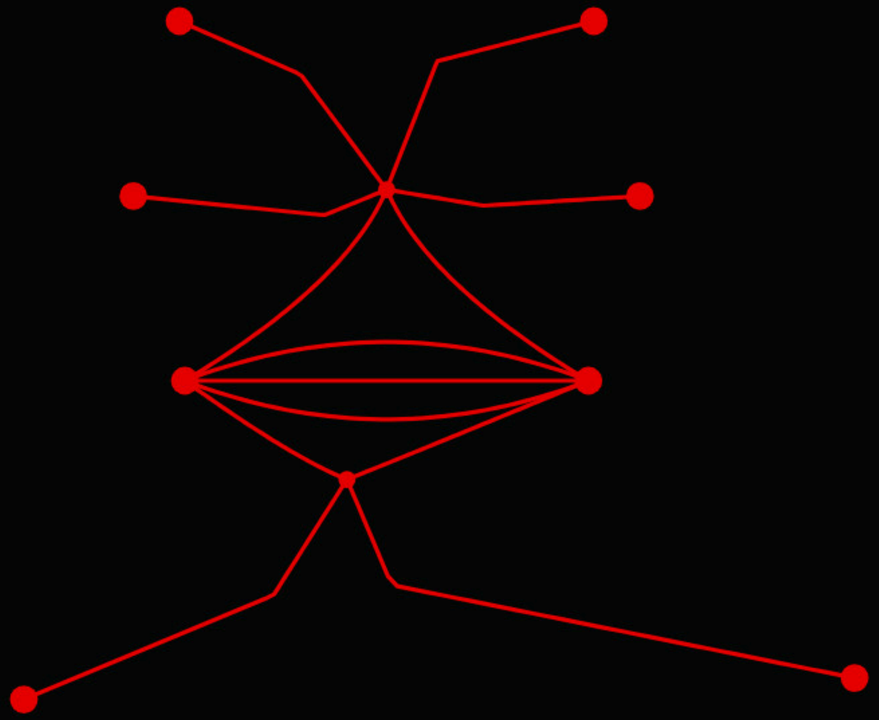
Dots and lines

Euler, 1736



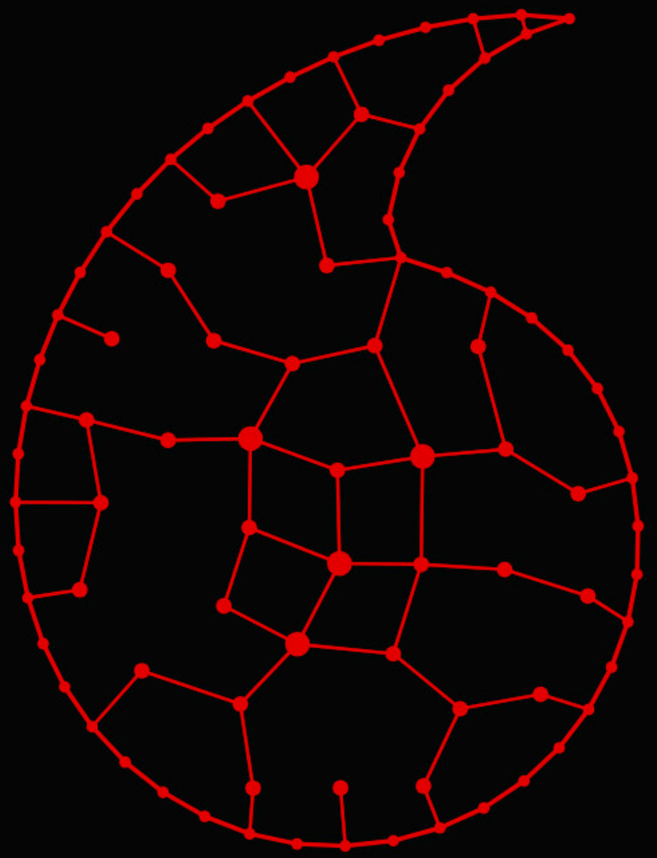
Dots and lines

Euler, 1736

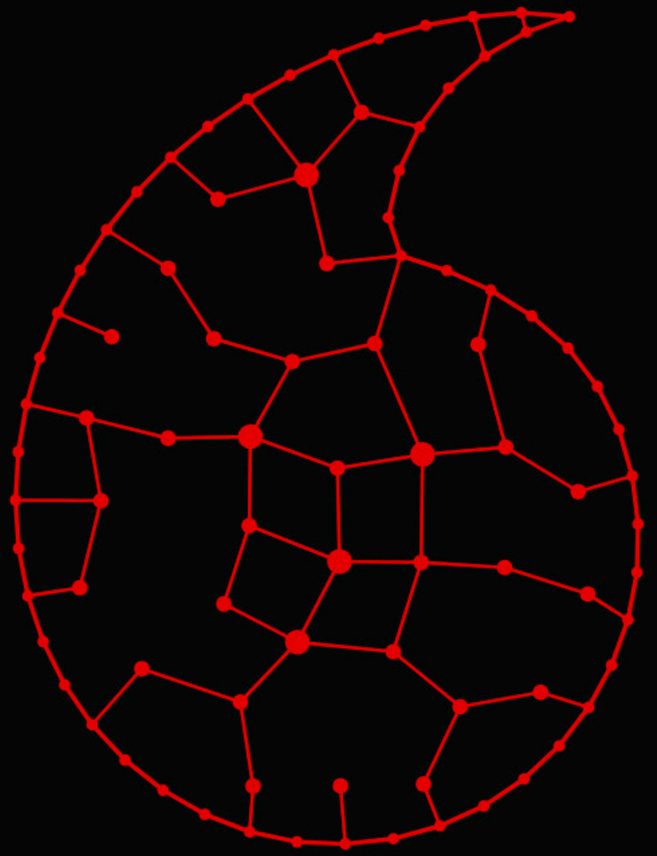


Sometimes the answer is the smaller part of what you find.

Together we can



Together we can



Together we can

Questions?